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AURA-MAS: Agentic and Multi-Agent Intelligent Surveillance

A Privacy-Aware, Edge-First Multi-Agent System for Multimodal Event Detection,

Inter-Sensor Coordination, and Explainable Alerting

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Abstract

Conventional video surveillance often sends many camera streams to a control room, where a small team must notice rare critical events. This thesis presents **AURA-MAS** (Agentic Understanding & Response Architecture), a privacy-aware, edge-first *hierarchical multi-agent system* for site surveillance. The system detects security-relevant events, combines related evidence, can request verification, and presents the result to a human operator.

At the edge, video agents use YOLO11 detection, ByteTrack tracking, zone rules, and optional CLIP-based anomaly scoring. Audio agents classify sound events and fall back to signal processing when required. They publish structured events rather than raw footage on an MQTT/Redis-compatible bus. A fusion agent groups events into hypotheses with a noisy-OR confidence score. For uncertain hypotheses, the coordinator can run a Contract Net-inspired auction to select a camera for verification. A deterministic policy agent applies thresholds, severity rules, and cooldowns. An explanation layer runs after the alert decision and checks every cited evidence identifier against the event log before presenting a report.

The Python implementation was evaluated in a 373-run campaign over nine annotated replay scenarios. Paired Wilcoxon tests with Holm–Bonferroni correction found no statistically significant event-level F1 difference among the four architectures. The auction is implemented, but its field-of-view term was inactive during measurement, so the campaign does not validate the intended spatial allocation. Cross-modal fusion occurs, but repeated evidence from one camera saturates noisy-OR before its confidence effect can be isolated. The explanation guardrail rejects unsupported evidence references, and the privacy pipeline anonymizes exported evidence while keeping an append-only decision log. The thesis therefore places most confidence in its evaluation apparatus: isolated subprocesses, repeated runs, a preserved pre-fix baseline, and a written methodology-change log.

Keywords— Intelligent Surveillance, Multi-Agent Systems, Agentic AI, Large Language Models, Object Detection, Multimodal Fusion, Auction-Based Coordination, Contract Net Protocol, Edge Computing, Explainable AI, Privacy by Design, Video Anomaly Detection

Résumé

La vidéosurveillance conventionnelle demeure largement centralisée et passive : les flux de nombreuses caméras convergent vers une salle de contrôle où des opérateurs surchargés doivent remarquer des événements critiques rares, un paradigme qui passe mal à l'échelle et réagit lentement. Cette thèse propose et évalue **AURA-MAS**, un *système multi-agents hiérarchique*, respectueux de la vie privée et orienté *edge*, pour la surveillance intelligente de sites : il détecte, corrobore, vérifie et explique les événements de sécurité tout en maintenant l'opérateur humain au centre de la décision.

Au niveau de la périphérie, des agents de perception autonomes traitent la vidéo (détection YOLO11 avec suivi multi-objets ByteTrack, règles de zones déclaratives et score d'anomalie sémantique zero-shot fondé sur CLIP) et l'audio (classification d'événements sonores avec repli en traitement du signal), en publiant des événements structurés — jamais de flux bruts — sur un bus de messages léger MQTT/Redis. Au niveau de la coordination, un agent de fusion agrège les événements en hypothèses d'incident par fusion tardive spatio-temporelle avec combinaison de confiance noisy-OR ; un agent coordinateur résout les hypothèses incertaines par allocation de tâches aux enchères inspirée du protocole Contract Net, en chargeant la caméra la mieux placée de vérifier activement l'incident ; et un agent de politique déterministe applique des seuils auditables, une cartographie de sévérité et une hystérésis d'alerte. Une couche d'explication *agentique* encadrée par des règles, fondée sur les grands modèles de langage, transforme ensuite les alertes acceptées en rapports d'incident en langage naturel dont les citations de preuves sont vérifiées mécaniquement par rapport au journal d'événements réel.

Le système complet est implémenté en Python et évalué par une campagne de 373 exécutions sur neuf scénarios de jeu annotés. Des tests de Wilcoxon appariés avec correction de Holm-Bonferroni ne mettent en évidence aucune différence statistiquement significative entre les quatre architectures pour la F1 au niveau événement. Le mécanisme d'enchères est implémenté, mais son terme informatif de recouvrement de champ de vision était inactif durant la mesure : il n'est donc pas validé tel qu'il est spécifié. La fusion multimodale se déclenche, mais son effet pratique sur la confiance ne peut être isolé car des preuves répétées de la même caméra saturent le noisy-OR. Le garde-fou d'explication et le pipeline de protection de la vie privée constituent des contributions structurelles : le premier rejette les références de preuves non étayées, tandis que le second maintient l'anonymisation des preuves exportées et une journalisation des décisions en ajout seul. La contribution empirique la plus solide est ainsi l'appareil d'évaluation lui-même : isolation par sous-processus, répétitions, conservation d'une base pré-correction et journal écrit des changements méthodologiques.

Mots-clés : Surveillance intelligente, Systèmes multi-agents, IA agentique, Grands modèles de langage, Détection d'objets, Fusion multimodale, Coordination par enchères, Protocole Contract Net, Informatique en périphérie, IA explicable, Protection de la vie privée dès la conception, Détection d'anomalies vidéo

ملخص

لا تزال المراقبة بالفيديو التقليدية مركزية وسلبية إلى حد كبير: تتدفق بثوث كاميرات عديدة نحو غرفة تحكم يتعين فيها على مشغلين مثقلين ملاحظة أحداث حرجة نادرة، وهو نموذج ضعيف التوسع وبطيء الاستجابة. تقترح هذه الأطروحة وتقيم نظام AURA-MAS، وهو نظام متعدد الوكلاء هرمي يحترم الخصوصية ويعمل على حافة الشبكة، للمراقبة الذكية للمواقع: يكتشف الأحداث الأمنية ويعززها ويتحقق منها ويشرحها مع إبقاء المشغل البشري في قلب القرار.

على مستوى الحافة، يعالج وكلاء إدراك مستقلون الفيديو كشف YOLO11 مع تتبع متعدد الأجسام ByteTrack وقواعد مناطق تصريحية ودرجة شذوذ دلالية بلا أمثلة قائمة على CLIP والصوت تصنيف الأحداث الصوتية مع بديل قائم على معالجة الإشارة، وينشرون أحداثاً مهيكلية --- وليس تدفقات خام أبداً --- عبر ناقل رسائل خفيف MQTT/Redis. وعلى مستوى التنسيق، يجمع وكيل الدمج الأحداث في فرضيات حوادث عبر دمج متأخر زمني-مكاني، ويحل وكيل التنسيق الفرضيات غير المؤكدة عبر تخصيص المهام بالمزاد المستوحى من بروتوكول Contract Net، بتكليف الكاميرا الأنسب بالتحقق النشط من الحادث، بينما يطبق وكيل السياسة الحتمي عتبات قابلة للتدقيق وخريطة خطورة وتباطؤ الإنذارات. ثم تحوّل طبقة شرح وكيالية مفيدة بالقواعد ومبنية على النماذج اللغوية الكبيرة الإنذارات المقبولة إلى تقارير حوادث بلغة طبيعية تُفحص استشهاداتها بالأدلة آلياً مقابل سجل الأحداث الفعلي.

تُفذ النظام الكامل بلغة بايثون وقيم من خلال حملة تضم 373 تشغيلاً عبر تسعة سيناريوهات إعادة تشغيل مشروحة. ولا تُظهر اختبارات Wilcoxon المزدوجة مع تصحيح Holm-Bonferroni فروقاً ذات دلالة إحصائية بين البنى الأربع في مقياس F1 على مستوى الحدث. آلية المزاد منفذة، لكن حذّها المعلوماتي لتداخل مجال الرؤية كان غير نشط أثناء القياس، ومن ثم لم تُثبت صلاحيتها كما حُدّت. يعمل الدمج متعدد الوسائط، لكن لا يمكن عزل أثره العملي على الثقة لأن الأدلة المتكررة من الكاميرا نفسها تُشبع نموذج noisy-OR. وتمثل حماية شرح التنبيهات وخط أنابيب الخصوصية مساهمتين بنيويتين: الأول يرفض مراجع الأدلة غير المدعومة، والثاني يحافظ على إخفاء هوية الأدلة المصدّرة وسجل قرارات تراكمي. لذلك فإن أقوى مساهمة تجريبية هي جهاز التقييم نفسه: عزل العمليات الفرعية، والتكرارات، وخط أساس محفوظ قبل الإصلاح، وسجل مكتوب لتغيرات المنهجية.

الكلمات المفتاحية: المراقبة الذكية، الأنظمة متعددة الوكلاء، الذكاء الاصطناعي الوكيلي، النماذج اللغوية الكبيرة، كشف الأجسام، الدمج متعدد الوسائط، التنسيق بالمزاد، بروتوكول Contract Net، الحوسبة الطرفية، الذكاء الاصطناعي القابل للتفسير، الخصوصية منذ التصميم، كشف الشذوذ في الفيديو.

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Acronyms

- ACL** Agent Communication Language. 18
- AUC** Area Under the Curve. 56
- BDI** Belief–Desire–Intention. 18
- CLIP** Contrastive Language–Image Pre-training. 22, 30, 56
- CNP** Contract Net Protocol. 14, 18, 31
- DSP** Digital Signal Processing. 23, 31, 36, 48, 49
- FIPA** Foundation for Intelligent Physical Agents. 18
- FPS** Frames Per Second. 30, 42
- GDPR** General Data Protection Regulation. 15, 25
- IoT** Internet of Things. 18, 24
- IoU** Intersection over Union. 31, 56
- JSON** JavaScript Object Notation. 18, 30, 32, 33, 36, 37
- LLM** Large Language Model. 13, 14, 19, 20, 25, 28, 36, 50, 55
- mAP** mean Average Precision. 21, 41
- MARL** Multi-Agent Reinforcement Learning. 19, 24
- MAS** Multi-Agent System. 13, 14, 18, 19, 28, 58
- MQTT** Message Queuing Telemetry Transport. 14, 18, 24, 29, 54, 55
- QoS** Quality of Service. 24, 29, 33
- Re-ID** Person Re-Identification. 15, 33
- SED** Sound Event Detection. 22, 25
- VAD** Video Anomaly Detection. 13, 22, 23, 25
- VLM** Vision-Language Model. 13, 14, 20, 25, 28, 32
- YOLO** You Only Look Once. 13, 21

Part I

Introduction

Chapter 1

General Introduction

1.1 Context and Motivation

Video surveillance is widely used in cities, campuses, industrial facilities, and transport infrastructure. In many installations, camera streams still converge on a control room where a small team monitors many screens and must notice rare safety-critical events in real time. Sustained attention across multiple feeds declines quickly, and the number of cameras often makes continuous manual monitoring impossible. As a result, operators frequently review footage after an incident rather than preventing it.

Early intelligent-surveillance systems added motion detection, then object detection and video anomaly detection (Video Anomaly Detection (VAD)), to centralized servers that ingest every stream. Detection accuracy has improved; modern You Only Look Once (YOLO) models run in real time and weakly supervised VAD methods perform well on benchmarks such as UCF-Crime [1]. Centralization still concentrates bandwidth and compute, creates a single point of failure, and moves identifiable footage away from the camera. It can also give operators many alerts without enough context to act on them.

Two lines of work suggest an alternative. *Multi-agent systems* (Multi-Agent System (MAS)) distribute perception and decision-making across cooperating software agents. They also provide coordination mechanisms such as the Contract Net Protocol [2] and auction-based task allocation [3]. *Agentic AI* uses large language models (Large Language Model (LLM)s) and vision-language models (Vision-Language Model (VLM)s) to plan, call tools, and reason over multimodal evidence [4, 5]. In surveillance, such a layer can explain the evidence produced by the perception system without deciding whether an alert should be raised.

This thesis asks how a surveillance system should organize perception, coordination, alerting, and explanation while keeping privacy and operator oversight in view.

1.2 Problem Statement

The problem addressed in this thesis can be stated as follows:

Design, implement, and evaluate a multi-agent intelligent surveillance system for a semi-closed site, in which autonomous perception agents detect security-relevant events from video and audio at the edge; coordinate with one another to corroborate and actively verify uncertain detections; apply deterministic, auditable alerting policies; and generate grounded natural-language explanations for a human operator—all under privacy-by-design constraints.

This problem decomposes into five research questions:

- **RQ1 (Architecture).** Does a distributed, hierarchical MAS architecture measurably improve responsiveness (time-to-alert) and scalability over a centralized sequential pipeline processing the same sensor streams?
- **RQ2 (Coordination).** Does explicit inter-agent coordination—specifically auction-based verification task allocation inspired by the Contract Net Protocol (CNP)—improve alert precision compared to uncoordinated per-sensor alerting, and at what communication cost?
- **RQ3 (Multimodality).** Does fusing audio events with video events increase detection confidence and reduce false alerts compared to vision-only operation?
- **RQ4 (Agentic explanation).** Can an LLM-based agentic layer generate incident explanations that are strictly grounded in system evidence—i.e., provably free of fabricated references—without influencing or delaying the safety-critical alert decision?
- **RQ5 (Governance).** Can the entire pipeline operate under privacy-by-design constraints (no raw footage leaves the edge, all exported evidence anonymized, all decisions audited) while remaining useful to the operator?

1.3 Objectives and Contributions

The overall objective is to deliver a working, evaluated prototype—named **AURA-MAS** (Agentic Understanding & Response Architecture — Multi-Agent System)—together with a reproducible evaluation methodology. The contributions of this thesis are:

- C1 — A hierarchical multi-agent surveillance architecture.** A three-layer design (edge perception, site coordination, governance/operator) with six implemented agent roles plus an operator console, a formal event/alert message schema, and a lightweight Message Queuing Telemetry Transport (MQTT)/Redis communication substrate (Chapter 5).
- C2 — An auction-based active verification protocol.** A single-round auction mechanism, derived from the CNP, in which uncertain “gray-zone” incident hypotheses trigger a verification task auctioned among camera agents; the winning agent performs a high-scrutiny re-analysis whose outcome adjusts the fused confidence before the alert decision (Chapters 5–6).
- C3 — Multimodal spatio-temporal late fusion.** A noisy-OR confidence combination scheme with per-modality reliability weights and cross-modal corroboration bonuses, which guarantees that independent supporting evidence strictly increases incident confidence (Chapter 5).
- C4 — A rule-guarded agentic explanation layer.** An LLM/VLM agent that produces natural-language incident reports *after* the deterministic alert decision, with a mechanical guardrail that verifies every cited evidence identifier against the actual event log and falls back to a deterministic template on any violation (Chapter 6).
- C5 — A reproducible evaluation methodology and ablation study.** Scenario replay infrastructure with ground-truth manifests, system-level metrics (event-level F1, mean time-to-alert, false alerts per hour, coordination overhead), and a four-way architectural ablation: centralized baseline, MAS without coordination, MAS with rule-based scheduling, and MAS with auction coordination (Chapter 7).

Revision notice

The contributions are implementation claims, not blanket performance claims. The final repeated campaign finds no Holm-corrected significant F1 difference between the four architectures; in particular, the auction’s field-of-view term was not populated during measurement. Chapter 7 and Appendix A record the resulting qualification.

1.4 Scope and Assumptions

The thesis targets *semi-closed sites*—campuses, warehouses, industrial facilities—where the set of cameras and microphones is known, zones can be declared (restricted areas, entries), and a single security operator team is responsible. The system performs *event detection and alerting*, not person identification: no facial recognition, biometric matching, or person re-identification (Person Re-Identification (Re-ID)) across time is performed, by deliberate design choice aligned with the General Data Protection Regulation (GDPR) data-minimization principle [6] and the prohibitions and high-risk provisions of the EU AI Act [7]. Evaluation is performed by replaying recorded multi-sensor scenarios with ground-truth annotations rather than live deployment, which enables controlled, repeatable ablations.

1.5 Thesis Organization

The remainder of this thesis is organized as follows. Chapter 2 introduces the theoretical foundations of multi-agent systems and agentic AI. Chapter 3 covers the perception building blocks: object detection, multi-object tracking, video anomaly detection, and audio event detection. Chapter 4 surveys the state of the art in distributed and multi-agent surveillance, agentic orchestration frameworks, and the legal/ethical governance landscape, and positions our contribution. Chapter 5 presents the design of AURA-MAS: architecture, agent roles, message schemas, fusion and coordination mechanisms, and privacy-by-design measures. Chapter 6 details the implementation: technology stack, code organization, and the algorithms of each agent. Chapter 7 describes the experimental setup, metrics, results of the architectural ablation, and discusses findings and limitations. Chapter 8 concludes and outlines future work.

Part II

Background and State of the Art

Chapter 2

Multi-Agent Systems and Agentic AI: Foundations

2.1 Introduction

This chapter establishes the conceptual vocabulary used throughout the thesis. We first define agents and multi-agent systems, then review the coordination mechanisms most relevant to distributed surveillance—in particular task allocation by auction—and finally introduce the recent paradigm of *agentic AI*, in which large language models act as reasoning engines inside agent architectures.

2.2 Agents and Multi-Agent Systems

2.2.1 The Agent Abstraction

Following Wooldridge [8], an **agent** is a computer system situated in an environment, capable of *autonomous action* in that environment in order to meet its design objectives. Four properties are commonly required:

- **Autonomy:** the agent operates without direct external control of its internal state and decisions;
- **Reactivity:** it perceives its environment and responds in a timely fashion to changes;
- **Pro-activeness:** it exhibits goal-directed behavior, taking initiative rather than only reacting;
- **Social ability:** it interacts with other agents through some communication language or protocol.

In AURA-MAS, a camera perception agent exemplifies all four: it autonomously processes its own video stream (autonomy), emits events when zone rules fire (reactivity), continuously maintains tracking state and dwell-time counters (pro-activeness), and bids on verification tasks announced by a coordinator (social ability).

2.2.2 Agent Architectures

Classical agent architectures span a spectrum from purely *reactive* (stimulus–response rules, subsumption) to *deliberative* (symbolic planning over explicit world models), with the influential

Belief–Desire–Intention (Belief–Desire–Intention (BDI)) model [9] occupying a pragmatic middle ground: agents maintain *beliefs* about the world, adopt *desires* (goals), and commit to *intentions* (plans). Hybrid layered architectures combine a fast reactive layer with a slower deliberative layer—a pattern that AURA-MAS mirrors at the system level: edge agents are predominantly reactive (frame-rate perception loops), while the coordination layer deliberates over aggregated hypotheses at a slower cadence, and the agentic explanation layer performs the slowest, most expensive reasoning only when justified.

2.2.3 Multi-Agent Systems

A **multi-agent system** (MAS) is a collection of interacting agents that jointly solve problems beyond the capability or knowledge of any individual agent [8]. MAS are attractive for surveillance for several structural reasons:

- **Natural distribution:** sensors are physically distributed; assigning one agent per sensor matches the problem topology and keeps raw data local.
- **Scalability:** adding a camera means adding an agent, not re-dimensioning a central server.
- **Robustness:** the failure of one agent degrades coverage locally instead of collapsing the whole system.
- **Heterogeneity:** video, audio, and higher-level reasoning agents can coexist behind a common message interface.

Interaction is standardized by agent communication languages; the Foundation for Intelligent Physical Agents (FIPA) Agent Communication Language (ACL) defines performatives (**inform**, **request**, **cfp**, **propose**, **accept-proposal**, ...) whose semantics underpin most practical coordination protocols. AURA-MAS adopts the *spirit* of FIPA interaction protocols while using lightweight JavaScript Object Notation (JSON) messages over MQTT topics, a common engineering compromise in modern Internet of Things (IoT)-oriented deployments [10].

2.3 Coordination and Task Allocation

2.3.1 The Coordination Problem

Coordination is the process of managing dependencies between agents’ activities. In distributed surveillance the canonical dependency is *verification*: when one sensor produces an uncertain detection, which other sensor should spend resources to confirm or refute it? This is an instance of the *multi-robot/multi-agent task allocation* problem, for which three broad families of solutions exist.

2.3.2 The Contract Net Protocol and Auctions

The Contract Net Protocol (CNP) introduced by Smith [2] structures task allocation as a negotiation: a *manager* announces a task (call for proposals), *contractors* evaluate it against their local capabilities and submit *bids*, and the manager awards the task to the best bidder. Market-based extensions formalize bids as utilities and prove strong properties for single-item auctions [3]: allocation quality equals the best available bid, communication cost is linear in the number of bidders, and the mechanism degrades gracefully when agents fail to respond.

Auctions are particularly well suited to camera verification tasks because a camera’s utility for verifying an event is inherently *local*, *private knowledge*—its field-of-view overlap with the event zone, its current processing load, whether it originated the initial detection—which the bidder can

compute cheaply and truthfully summarize as a scalar bid. AURA-MAS therefore adopts a single-round, sealed-bid auction (Section 5.7) with a fixed bidding window, and retains a round-robin scheduler as an ablation baseline.

2.3.3 Task-Allocation Scope and Null Baselines

The allocation instance in this thesis is intentionally small: one uncertain hypothesis is offered to a set of cameras, one camera is selected, and the decision is immediate. It is therefore closer to a single-task, single-assignment allocation problem than to a continuous fleet-planning problem. This scope matters when interpreting a market mechanism. At a two-camera site, a deterministic round-robin or “other camera” rule is a strong and transparent null: a complex allocation protocol must exploit real information about viewpoint, load, or expected evidence to beat it. If that information is absent, using an auction changes message flow without proving a better decision.

For larger sites, assignment can be formulated as a utility matrix over tasks and cameras and solved by a Hungarian-style centralized matcher, or by distributed bundle methods. Neither is implemented here. They are useful future baselines precisely because they distinguish the question “should a camera verify?” from “which camera is best placed to verify?” The campaign audit in Chapter 7 shows why this distinction cannot be assumed from the presence of an auction alone.

2.3.4 Alternative Approaches

Two alternatives deserve mention. *Centralized optimization* (e.g., Hungarian assignment over a global utility matrix) yields optimal allocations but requires the center to know all utilities, reintroducing the communication and single-point-of-failure costs that MAS avoid; recent work explores submodular optimization for multi-camera collaborative perception [11]. *Multi-agent reinforcement learning* (Multi-Agent Reinforcement Learning (MARL)) learns coordination policies end-to-end, with algorithms such as QMIX [12] and MADDPG [13] demonstrating strong results in simulated cooperative sensing and patrolling [14, 15]; standardized environments (PettingZoo [16]) and scalable training stacks (RLlib [17]) exist. However, MARL policies are data-hungry, hard to certify, and opaque—undesirable properties for a safety- and legally-sensitive system. We therefore position MARL-based active sensing as future work (Chapter 8) and keep the deployed coordination mechanism auditable.

2.4 Agentic AI

2.4.1 From Language Models to Agents

Large language models exhibit emergent capabilities for planning, tool use, and multi-step reasoning. The *agentic AI* paradigm operationalizes these capabilities: an LLM is embedded in a control loop where it observes state, reasons (often via explicit “thought” traces as in ReAct [5]), invokes tools, and iterates. Sapkota et al. [4] distinguish *AI agents*—single-model, tool-using systems—from *agentic AI*: orchestrated ensembles of specialized agents with shared memory and explicit coordination, which is precisely the regime of interest for surveillance, where perception specialists must feed reasoning specialists.

2.4.2 Orchestration Frameworks

A rich ecosystem of frameworks now supports multi-agent LLM orchestration: AutoGen structures multi-agent conversations [18]; LangGraph models agent workflows as typed state graphs with explicit nodes, edges, and checkpointing [19]; CrewAI emphasizes role-based teams. For

vision-centric pipelines, research systems such as VideoAgent [20] and Hawk [21] demonstrate LLM-orchestrated video understanding, while VLMs such as GPT-4V [22], Gemini [23], Qwen-VL [24], and LLaVA [25] provide the underlying multimodal perception-to-language capability.

2.4.3 Reliability and Guardrails

The central obstacle to deploying generative models in safety-critical loops is *hallucination*: fluent but unfounded output. Guardrail frameworks such as NeMo Guardrails [26] interpose programmable checks between the model and the application. The design stance adopted in this thesis is stricter and architectural: the agentic layer is *decision-decoupled*—it runs only after a deterministic policy engine has already made the alert decision—and *evidence-grounded*—every evidence identifier cited in a generated report is mechanically checked against the event log, with automatic fallback to a template on violation. This turns the LLM from a potential safety liability into a strictly additive usability layer, a pattern we consider one of the transferable lessons of this work.

2.5 Conclusion

This chapter introduced agents, multi-agent coordination, and agentic AI. It explains why AURA-MAS uses an auction for verification and keeps the explanation layer outside the alert decision. The next chapter reviews the perception methods used at the edge.

Chapter 3

Perception for Intelligent Surveillance

3.1 Introduction

Every surveillance decision ultimately rests on perception: converting raw pixels and audio samples into structured, semantically meaningful events. This chapter reviews the four perception capabilities used by AURA-MAS edge agents—object detection, multi-object tracking, video anomaly detection, and audio event detection—with emphasis on the accuracy/latency trade-offs that govern edge deployment.

3.2 Object Detection

Object detection localizes and classifies objects in images. Modern real-time detection is dominated by the YOLO family, which frames detection as a single dense regression pass over the image. The current generation (YOLO11, released by Ultralytics in 2024 [27]) provides a spectrum of model scales; the nano variant (YOLO11n, $\approx 2.6\text{M}$ parameters) achieves approximately 39.5% mean Average Precision (mAP) on COCO while sustaining real-time throughput on CPU and embedded hardware, making it the natural choice for per-camera edge agents. Transformer-based detectors such as RT-DETR [28] eliminate non-maximum suppression and reach higher accuracy at moderate additional cost; they constitute a drop-in upgrade path in our architecture since agents encapsulate their detector behind a common event interface.

Two practical considerations matter for surveillance specifically. First, detection confidence thresholds trade recall against precision at the *event* level, not just the box level: a missed person is a missed intrusion. Second, pretrained COCO classes (person, car, truck, backpack, ...) cover the vast majority of surveillance-relevant objects, so fine-tuning is optional rather than blocking—an important de-risking property for a time-boxed project [29].

For zone-based rules, a bounding box is not itself a location model. AURA-MAS uses the bottom centre of a detected box (the foot point) and a ray-casting test against the declared polygon. This is a practical image-plane heuristic: it is often more stable than the box centre for a standing person, but it can fail under occlusion, a steep top-down view, a cropped body, or a detector box that includes substantial background. The IoU used for static-object logic is defined by

$$\text{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|}.$$

An IoU threshold of 0.6 has different operational meaning for a small backpack than for a truck-sized box; scale, detector jitter, and camera perspective change the same numerical overlap. This is why the thresholds in Appendix D are disclosed as implementation parameters rather than universal surveillance constants.

3.3 Multi-Object Tracking

Detection alone yields per-frame boxes; surveillance rules (loitering, line crossing, abandonment) require *identity over time*. Multi-object tracking (MOT) associates detections across frames into trajectories. The prevailing paradigm is tracking-by-detection, and the reference algorithm is ByteTrack [30], whose key insight is to associate *every* detection box—including low-confidence ones—in a two-stage matching (high-confidence first, then low-confidence against unmatched tracks), which markedly reduces identity switches under occlusion. ByteTrack is integrated natively in the Ultralytics runtime (`model.track(..., tracker="bytetrack.yaml")`), which AURA-MAS uses directly. Evaluation of trackers uses HOTA/MOTA/IDF1 metrics computed by tools such as TrackEval [31, 32]; multi-camera extensions of tracking (cross-camera re-identification) exist [33, 34] but are deliberately excluded from this thesis on privacy grounds—our cross-camera corroboration operates at the *event* level, not the identity level.

ByteTrack’s second association stage is especially relevant at the deployed 5-FPS sampling rate: low-confidence detections can bridge a short occlusion, but sparse temporal sampling gives an object more time to move between frames and therefore makes association harder. The corpus contains no per-frame track annotations, so HOTA, MOTA, and IDF1 are named here as appropriate tracker metrics but are deliberately not reported as results. A field-of-view homography would provide a more principled relation between camera views and ground-plane zones than image-plane polygons; it is also the missing input needed to make the auction’s overlap term meaningful.

3.4 Video Anomaly Detection

VAD aims to flag frames or segments that deviate from normal patterns. Three families coexist [35]:

- **Reconstruction/prediction methods** (autoencoders, future-frame prediction) trained on normal data only; effective on constrained scenes (Avenue, ShanghaiTech) but brittle under domain shift.
- **Weakly-supervised methods**, trained with video-level labels on datasets such as UCF-Crime [1]; the current accuracy leaders.
- **Zero-shot semantic methods**, which exploit vision-language alignment: a frame embedding is compared against natural-language descriptions of normal versus anomalous situations using Contrastive Language–Image Pre-training (CLIP) [36]. Recent variants (VadCLIP-style, AVadCLIP with audio [37], training-free EventVAD [38], explainable EX-VAD [39]) show that useful anomaly signals can be obtained *without any training*, which is exceptionally attractive for a one-month project.

AURA-MAS uses the third family as an optional semantic scorer on sampled frames: anomaly score is the softmax mass assigned to a set of anomalous text prompts (“people fighting”, “fire and smoke”, ...) relative to normal prompts. This complements—rather than replaces—the deterministic zone-rule engine, and every CLIP-triggered event is subject to the same downstream fusion, verification, and policy thresholds as any other event.

3.5 Audio Event Detection

Audio is a cheap, omnidirectional, lighting-invariant modality that video-only systems ignore. Sound event detection (Sound Event Detection (SED)) classifies acoustic events—glass breaking,

screams, gunshots, alarms—from short audio windows. Pretrained models dominate practice: YAMNet (MobileNet-based, 521 AudioSet classes) [40, 41], PANNs [42], the Audio Spectrogram Transformer [43], and BEATs [44] offer increasing accuracy at increasing cost; efficient distillations exist for edge deployment [45, 46]. A surveillance-relevant subset of classes can be mapped directly to alert-worthy event types, with per-class confidence thresholds. Because deep audio stacks can be heavy for constrained hardware, AURA-MAS additionally implements a dependency-light Digital Signal Processing (DSP) fallback: rolling z-scores of short-time energy and spectral flatness flag acoustic transients as generic `audio_anomaly` events—sufficient to demonstrate multimodal fusion even where TensorFlow is unavailable.

The audio repair documented for the v2 campaign illustrates why temporal pre-processing is part of the method, not incidental plumbing. Audio is represented as a log-mel time–frequency signal for model inference. A one-second chunk is evaluated with a lookback extension and YAMNet scores are max-pooled across the model’s internal frames; using mean pooling on independently re-windowed one-second chunks diluted brief transients below threshold. The before/after observation and the campaign implications are recorded in Appendix E.

3.6 Multimodal Fusion

Fusion strategies are classified as early (feature-level), late (decision-level), or hybrid. For a distributed system, *late fusion* is the natural fit: each agent emits calibrated event confidences, and a fusion node combines them without needing access to raw features—preserving both bandwidth and privacy. AURA-MAS groups events into spatio-temporal hypotheses (same incident family, same zone, within a sliding window) and combines confidences with a noisy-OR model weighted by per-modality reliability, plus small calibrated bonuses for cross-modality and cross-sensor corroboration (Section 5.6). Audio-visual VAD research confirms that such corroboration materially improves robustness [37, 47].

Noisy-OR is interpretable, but its usual probabilistic reading assumes that the contributing evidence is conditionally independent. Repeated events from one camera watching the same subject violate that assumption. In the present implementation they can still drive the product toward one, so monotonicity should not be conflated with calibration. The fusion audit in Chapter 7 treats this saturation as a limitation rather than as a multimodal gain.

3.7 Datasets and Evaluation Data

Public datasets relevant to this work include UCF-Crime (1,900 untrimmed real-world videos across 13 anomaly classes) [1], Avenue and ShanghaiTech for scene-specific VAD, VIRAT for activity detection from static surveillance cameras, and MEVA for large-scale multi-camera activity detection [48]. Synthetic data generation is an increasingly credible complement [49, 50, 51]. For *system-level* evaluation (as opposed to model-level), no standard benchmark exists that combines multi-sensor streams with ground-truth incident annotations; this thesis therefore contributes replayable scenario manifests built from public clips and scripted recordings (Chapter 7), following the methodological lead of ETISEO [52] and modern MLOps evaluation practice [53].

3.8 Conclusion

Pretrained detectors, trackers, zero-shot anomaly scorers, and audio classifiers can be combined without training a model for this project. The remaining challenge is how to fuse, verify, decide on, and explain their outputs. The rest of the thesis describes that architecture.

Chapter 4

State of the Art in Distributed and Agentic Surveillance

4.1 Introduction

This chapter reviews the research areas used in AURA-MAS: distributed and edge video analytics, multi-agent and MARL-based surveillance, agentic multimodal alerting, and privacy and legal governance. It then positions the prototype against the gaps identified in this review.

4.2 Distributed and Edge Video Analytics

Bandwidth, latency, and privacy have moved more video analysis from cloud services to the edge [54, 55]. NVIDIA DeepStream [56] shows multi-stream, hardware-accelerated inference at the edge. ONNX Runtime [57] and single-board computers make per-camera inference more practical [58]. Edge-cloud designs often filter easy cases locally and send ambiguous cases upward [59]. AURA-MAS applies the same idea across agents: edge perception, site coordination, then explanation. Distributed-surveillance studies also report benefits in elasticity and fault containment [60, 61].

On the communication substrate, comparative studies of IoT messaging protocols consistently find MQTT superior for constrained, event-driven workloads [62, 63, 64], with Mosquitto as the standard broker [10] and Quality of Service (QoS) levels providing per-topic delivery guarantees. Real-time alert pipelines increasingly pair a pub/sub bus with a durable stream store (Redis Streams / Kafka) for alerts and audit [65, 66, 67]. AURA-MAS adopts exactly this two-tier substrate.

4.3 Multi-Agent and Learning-Based Surveillance

Classical multi-camera coordination research addresses camera hand-off, field-of-view negotiation, and distributed tracking [33, 68]. Task allocation via market mechanisms remains the practical workhorse [2, 3], with modern variants addressing task-oriented communication under bandwidth constraints [69] and submodular sensor selection [11].

MARL is also used for cooperative sensing. Multi-UAV search [14], target coverage [15], and related coverage formulations [70] show that learned policies can outperform hand-crafted coordination in simulation. QMIX [12] and MADDPG [13] are common algorithms, with PettingZoo [16] and RLlib [17] supporting experimentation. Real surveillance deployments remain scarce. Sample complexity, sim-to-real transfer, and limited auditability make these methods difficult to use in safety-critical settings. AURA-MAS therefore uses an auction that can be inspected directly.

4.4 Agentic and Multimodal Alerting

Recent work applies LLM/VLM agents to video understanding. Hawk [21] and VideoAgent [20] use an LLM planner to orchestrate perception tools. AVadCLIP [37] and related methods [47] combine audio and video for anomaly detection. Other work uses description-based SED [71] or explainable VAD [39, 72] to make outputs easier to interpret. AutoGen [18] and LangGraph [19] provide orchestration machinery, while [4, 73, 74] review multi-agent LLM frameworks. Guardrails [26] and LLM-as-judge evaluation [75] can reduce hallucination, but do not remove it. In the systems reviewed here, the generative component is not architecturally separated from the alert decision; it is either part of detection or an unchecked narrator.

4.5 Privacy, Ethics, and Legal Governance

Surveillance AI is now heavily regulated territory. The GDPR imposes lawfulness, data minimization, and storage limitation on video processing [6]; European guidance discourages centralized retention of identifiable footage [76, 77]. The EU AI Act [7] prohibits real-time remote biometric identification in public spaces (with narrow exceptions) and classifies most other surveillance AI as high-risk, mandating risk management, logging, human oversight, and transparency [78, 79, 80, 81]. Trustworthy-AI frameworks (NIST AI RMF [82], reviewed in [83, 84]) converge on the same operational requirements: edge-local processing, anonymization, audit trails, and human-in-the-loop decision authority. These requirements are *constitutive* design constraints for AURA-MAS, not afterthoughts.

4.6 Synthesis and Positioning

Table 4.1 summarizes the gap analysis.

Table 4.1: Gap analysis: existing system families versus AURA-MAS.

Capability	Centralized VMS analytics	MARL surveillance (research)	LLM video agents	AURA-MAS (this thesis)
Edge-local perception	Partial	Simulated	No (cloud VLM)	Yes
Explicit coordination	No	Learned, opaque	Conversational	Auction (auditable)
Multimodal fusion	Rare	Rare	Partial	Yes (noisy-OR late fusion)
Explainable alerts	No	No	Yes, ungrounded	Yes, evidence-checked
Decision/generation decoupling	—	—	No	Yes (policy-first)
Privacy by design	Weak	Not addressed	Not addressed	Constitutive
System-level evaluation	Vendor claims	Simulation reward	Qualitative	Replayable ablations

This review identifies three gaps. **(G1)** None of the surveyed systems combines auditable multi-agent coordination with modern zero-shot perception. **(G2)** The surveyed agentic explanation

layers are not separated from decision-making and grounded by construction. **(G3)** Academic work usually evaluates models rather than complete systems with time-to-alert, false alerts per hour, and coordination overhead under a controlled ablation. AURA-MAS addresses these points at prototype scale and within explicit legal constraints.

4.7 Conclusion

The state of the art provides mature components and clear regulatory constraints but no integrated, auditable, privacy-preserving multi-agent architecture with grounded agentic explanation. The next chapter presents our design.

Part III

Contribution: Design of AURA-MAS

Chapter 5

AURA-MAS: System Design

5.1 Introduction

This chapter presents the design of AURA-MAS, from global architecture down to message schemas and coordination protocols. The design is governed by five principles derived from the analysis of Chapters 2–4:

1. **Edge-first:** raw sensor data is processed where it is produced and never leaves the edge agent;
2. **Events, not streams:** all inter-agent communication consists of small structured messages;
3. **Deterministic decisions:** alerts are produced exclusively by an auditable rule-based policy engine;
4. **Guarded generativity:** LLM/VLM components explain decisions but can never make or alter them;
5. **Privacy by design:** anonymization, minimization, and auditability are structural, not optional.

5.2 Global Architecture

AURA-MAS is a hierarchical MAS organized in three layers connected by a two-tier message substrate (Figure 5.1).

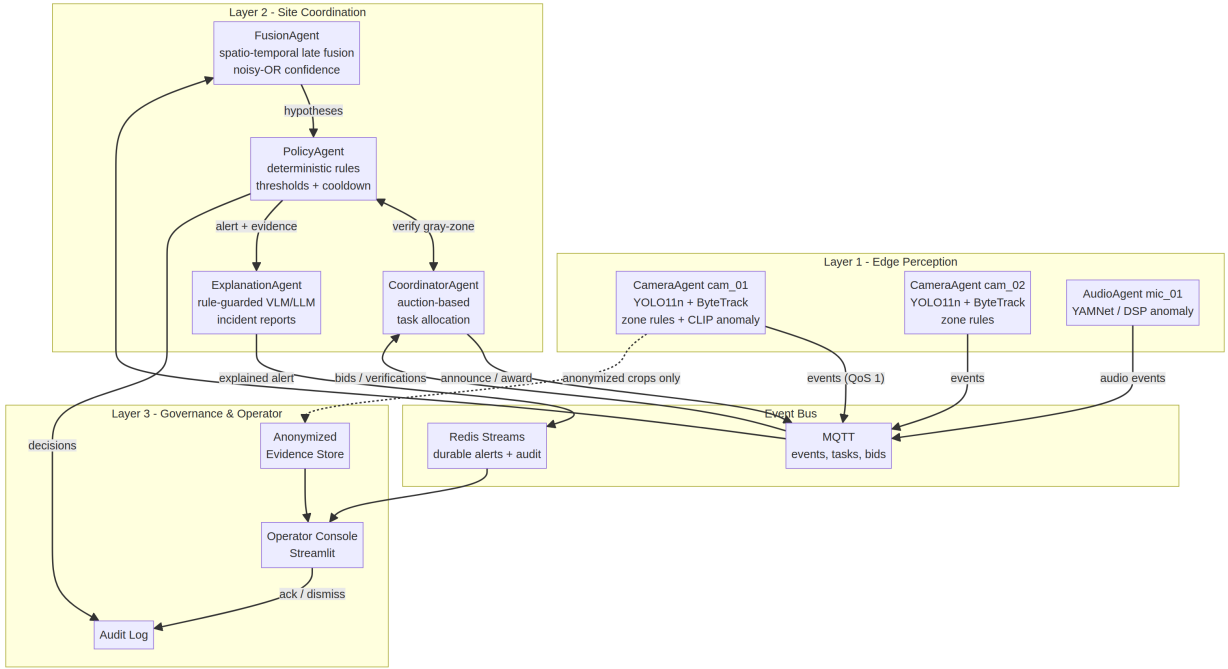


Figure 5.1: AURA-MAS architecture: edge perception layer, site coordination layer, and governance/operator layer, connected by an MQTT event bus and a Redis durable stream.

Layer 1: edge perception. One agent handles each sensor. *CameraAgents* run detection, tracking, zone rules, and optional semantic anomaly scoring on their video stream. *AudioAgents* classify sound events or score acoustic anomalies. Both publish structured events on the bus.

Layer 2: site coordination. The *FusionAgent* groups events into incident hypotheses. The *CoordinatorAgent* allocates verification tasks when a hypothesis is uncertain. The *PolicyAgent* applies deterministic alert or suppression rules. The *ExplanationAgent* produces a report for an accepted alert.

Layer 3: governance and operator. A web console shows alerts, anonymized evidence, and explanations. Operator actions and policy decisions are appended to the audit log. The log is append-only by convention, not cryptographically immutable.

The substrate pairs MQTT [10] for low-latency fan-out of high-frequency messages (detections QoS 0; events, tasks, bids, awards QoS 1) with Redis Streams for durable, replayable storage of alerts and audit records. A single-process in-memory bus with identical semantics supports development and testing.

5.3 Agent Roles and Responsibilities

Table 5.1: Agent roles in AURA-MAS.

Agent	Responsibility	Consumes	Produces
CameraAgent	Detection, tracking, zone rules, CLIP anomaly, verification bids/execution	video frames; tasks, awards	detections, events, bids, verifications
AudioAgent	Sound event classification / DSP anomaly	audio chunks	events
FusionAgent	Spatio-temporal hypothesis formation, confidence fusion	events	hypotheses
CoordinatorAgent	Auction-based verification task allocation	hypotheses (gray-zone), bids, verifications	tasks, awards
PolicyAgent	Severity mapping, thresholds, hysteresis, alert emission, audit	hypotheses, verification results	alerts, audit records
ExplanationAgent	Grounded incident report generation with guardrails	alerts, hypotheses, evidence	explanations
Operator console	Human oversight, acknowledgment	alerts, evidence	operator audit records

5.4 Message Schemas

All messages are JSON documents with mandatory provenance fields. The three core schemas are:

Detection (high-frequency, transient): {sensor_id, frame_id, timestamp, objects: [{class, confidence, bbox, track_id}]}.

Event (semantic, durable): {event_id, sensor_id, timestamp, event_type, confidence $\in [0,1]$, modality, zone, track_id, evidence_path, extra}. Event types include intrusion, loitering, abandoned_object, anomaly, and audio types (audio_glass_break, audio_scream, ...). The evidence_path always references an *anonymized* crop.

Alert (operator-facing): {alert_id, timestamp, severity $\in \{\text{CRITICAL}, \text{WARNING}, \text{INFO}\}$, event_type, confidence, zone, sensors[], evidence[], fused_events[], explanation, status}.

5.5 Edge Perception Design

5.5.1 Camera Pipeline

Each CameraAgent samples its stream at a configurable inference rate (default 5 Frames Per Second (FPS)) and runs a four-stage pipeline: (1) YOLO11n detection with ByteTrack association [27, 30]; (2) a stateful *zone rule engine* evaluating declarative site rules; (3) optional CLIP zero-shot anomaly scoring on subsampled frames [36]; (4) event emission with anonymized evidence.

The zone rule engine implements three rules over tracked objects, using the foot point (bottom-center of the bounding box) for zone membership tested by ray-casting against declared polygons:

- **Intrusion:** a person track enters a zone declared **restricted**; fired once per (track, zone) pair.
- **Loitering:** dwell time of a person track inside any zone exceeds a threshold τ_{loiter} (default 8 s); confidence grows with dwell time.
- **Abandoned object:** a non-person track remains static (Intersection over Union (IoU) with its initial box > 0.6) longer than τ_{aband} (default 10 s).

5.5.2 Audio Pipeline

AudioAgents process audio in overlapping windows extending up to 1.5 s of lookback context per 1-second chunk, so that short transient sounds are scored with sufficient temporal context rather than against a bare 1 s window. In YAMNet mode¹ [40], a curated mapping from AudioSet classes to surveillance event types (with per-class thresholds) produces typed events. In DSP fallback mode, rolling z-scores of short-time energy and spectral flatness produce generic **audio_anomaly** events, normalized to $[0, 1]$.

5.6 Fusion Design

The FusionAgent maintains a set of open *hypotheses*, keyed by (incident family, zone), where families group mutually corroborating event types (e.g., **intrusion** and **audio_glass_break** both belong to *security*). An incoming event either joins the open hypothesis for its key (if within the sliding window W , default 6 s, of the last event) or opens a new one.

Hypothesis confidence uses a reliability-weighted noisy-OR:

$$C(H) = 1 - \prod_{e \in H} (1 - w_{m(e)} \cdot c_e) + \beta_{mod} \cdot \mathbb{K}[|M(H)| > 1] + \beta_{sen} \cdot \mathbb{K}[|S(H)| > 1] \quad (5.1)$$

where c_e is event confidence, w_m a per-modality reliability weight (video 0.9, audio 0.7), $M(H)$ and $S(H)$ the sets of modalities and sensors contributing to H , and $\beta_{mod} = \beta_{sen} = 0.05$ small corroboration bonuses; the result is clamped to $[0, 1]$. The noisy-OR form guarantees monotonicity: additional supporting evidence never decreases confidence, and independent corroboration from a second modality or sensor strictly increases it—the formal core of contribution C3. When a hypothesis window closes, the hypothesis is flushed to the PolicyAgent.

5.7 Coordination Design

Fused confidence in the *gray zone* $[\theta_{lo}, \theta_{hi}]$ (defaults 0.35–0.75) means the system is neither confident enough to alert nor to dismiss. Rather than guessing, the CoordinatorAgent buys information through a single-round sealed-bid auction derived from the CNP [2]:

1. **Announce:** the coordinator publishes a verification task $\langle \text{task_id}, \text{event_type}, \text{zone}, \text{origin_sensor} \rangle$ on the task topic.
2. **Bid:** each CameraAgent computes a private utility $u = b \cdot \kappa \cdot o$ where b penalizes the origin sensor (0.3 vs 1.0, favoring independent confirmation), κ reflects current load (0.2 if busy), and $o \in [0, 1]$ is its field-of-view overlap with the event zone; it publishes the bid within the bidding window T_b (default 1 s).

¹The DSP fallback triggers not only when `tensorflow/tensorflow_hub` are absent, but also transiently on model-load failure; an earlier silent `except Exception` around the YAMNet load masked a dead model URL for an unknown period, motivating an explicit `backend="yamnet"` configuration option that now fails loudly instead of silently falling back.

3. **Award**: the coordinator awards the task to the highest bidder.
4. **Verify**: the winner re-analyzes its current frame at higher scrutiny (larger input resolution, lower confidence floor) and reports **verified** with a verification score.
5. **Integrate**: the PolicyAgent adjusts hypothesis confidence: $+0.15$ if verified, -0.20 if refuted.

The protocol costs $O(n)$ messages per task (n = number of cameras) and bounded wall time $T_b + T_{verify}$. A round-robin allocator (ignoring utilities) serves as the ablation baseline, and coordination can be disabled entirely.

Revision notice

The utility expression states the intended design, but the v2 campaign did not populate `fov_overlap` in its manifests. During those measurements, *o* consequently used the 0.5 fallback for every camera. The reported campaign does not validate spatially informed bidding; the correction and its evidence are discussed in Section 7.4.

5.8 Policy and Alerting Design

The PolicyAgent is the sole component authorized to create alerts. Its decision procedure is fully deterministic: (1) map the hypothesis’s dominant event type to a severity class (e.g., **intrusion** and **audio_gunshot** $\rightarrow$ CRITICAL; **loitering** $\rightarrow$ INFO); (2) optionally request verification for gray-zone confidence; (3) compare adjusted confidence to a per-severity threshold (CRITICAL 0.45, WARNING 0.55, INFO 0.70—lower thresholds for higher-stakes events reflect asymmetric miss costs); (4) apply per-(zone, type) cooldown (default 20 s) to prevent alert storms; (5) write *every* decision—alert or suppression, with reason—to the audit stream. This design directly implements the human-oversight and logging obligations of high-risk AI systems under the EU AI Act [7].

5.9 Agentic Explanation Design

The ExplanationAgent runs a small state machine (collect evidence $\rightarrow$ describe frames with a VLM $\rightarrow$ draft structured report $\rightarrow$ guardrail check), implementable in LangGraph [19] or directly. The system prompt constrains the model to use only supplied evidence, cite evidence by exact identifiers, avoid identity speculation, and recommend only observation actions. The *guardrail* then mechanically verifies that (a) the output parses to the required JSON schema and (b) every cited evidence identifier—in the citation list and in free text—exists in the hypothesis event log. Any violation triggers fallback to a deterministic template that is always available. Because the alert has already been emitted when explanation begins, generation latency and failures affect only report richness, never safety (contribution C4).

5.10 Privacy and Governance by Design

Table 5.2 maps regulatory requirements to design mechanisms.

Table 5.2: Privacy-by-design mechanisms in AURA-MAS.

Requirement (GDPR / AI Act)	Mechanism
Data minimization	Only JSON events cross the network; raw frames never leave the edge agent
No biometric identification	No face recognition or Re-ID; corroboration at event level only
Anonymization	All exported evidence passes through person/face blurring; fail-closed (full-frame blur if localization unavailable)
Storage limitation	Evidence retention window configurable; detections are transient (QoS 0, not persisted)
Human oversight	Operator console with acknowledge/dismiss authority; no automated enforcement actions
Logging & traceability	Append-only audit stream of every policy decision and operator action; cryptographic immutability is not implemented
Transparency	Grounded natural-language explanations with verifiable evidence citations

5.11 Conclusion

AURA-MAS assigns perception to sensor-local agents, uses an auction for uncertain verification, keeps alert decisions in a deterministic policy engine, and places generative AI after the decision. The next chapter describes the implementation.

Part IV

Implementation

Chapter 6

Implementation

6.1 Introduction

This chapter describes the concrete realization of the AURA-MAS design as an open, tested Python codebase with 4,376 tracked Python lines (5,188 when uncommitted working-tree files are included). The evaluated configuration is the tracked code; the larger count is reported only to make the repository state explicit. We present the technology stack, the repository organization, the key algorithms of each component, and the engineering decisions that keep the system runnable on commodity hardware without any model training.

6.2 Technology Stack

Table 6.1: Technology stack of the AURA-MAS prototype.

Concern	Technology	Rationale
Language	Python 3.11+	Ecosystem coverage of all required libraries
Detection + tracking	Ultralytics YOLO11n + ByteTrack [27, 30]	Real-time on CPU; tracking integrated
Semantic anomaly	OpenAI CLIP ViT-B/32 [36]	Zero-shot; no training required
Audio	YAMNet [40] / librosa DSP fallback	Pretrained 521-class SED; light fallback
Event bus	MQTT (Mosquitto, paho-mqtt) [10]	Low-latency pub/sub, QoS, wildcards
Durable store	Redis Streams	Replayable alert/audit log
Explanation	OpenAI-compatible chat API	Works with GPT-4o-mini, Qwen-VL, local Ollama
Dashboard	Streamlit and Next.js console	Operator-facing alert views for prototyping and wall-display use
Testing	pytest	Offline tests without models
Deployment	Docker Compose	One-command broker setup

The bus is hidden behind a small interface, `publish` and `subscribe` with MQTT-style wildcards. `LocalBus` implements the same interface, so the system and the auction can run in one process

without external services. Heavy dependencies also have fallbacks: the audio agent uses DSP when TensorFlow is unavailable, the camera agent can run rules without CLIP, and the explanation agent uses a template without an LLM endpoint. These choices keep the demonstrator runnable on a machine without the full production stack.

6.3 Repository Organization

```
aura_mas/
  core/      bus.py      (transports, Detection/Event/Alert schemas,
                        AlertStore with Redis/JSONL backends)
            privacy.py  (anonymization: person blurring, fail-closed)
  agents/    base.py      camera_agent.py  audio_agent.py
            fusion_agent.py  coordinator_agent.py
            policy_agent.py  explanation_agent.py
  scenarios/ replay.py    (scenario runner + centralized baseline)
  eval/      metrics.py   (event-level F1, time-to-alert, FAR/h, overhead)
  dashboard/ app.py        (Streamlit operator console)
  scripts/   make_synthetic_clips.py  probe_zones.py  make_figures.py
  tests/     offline unit and integration tests (26 tests; model-free subset available)
  frontend/  Next.js operator console
  scenarios/*.json  docker-compose.yml  requirements.txt
```

6.4 Core Components

6.4.1 Message Bus and Schemas

`core/bus.py` defines the three dataclass schemas (Detection, Event, Alert) with JSON serialization, topic constants, and three interchangeable transports: `MqttBus` (paho-mqtt), `LocalBus` (thread-safe in-process pub/sub with `+/#` wildcard matching), and an `AlertStore` that appends alerts and audit records to Redis Streams when available, mirroring to JSONL files otherwise. Topic layout: `site/{sensor}/detections`, `site/events`, `site/coord/{tasks,bids,awards,verifications}`, `site/alerts`.

6.4.2 Privacy Module

`core/privacy.py` exposes a single choke point, `anonymize_and_save(frame, boxes)`, through which *all* visual evidence must pass: person regions (YOLO boxes passed by the camera agent, or HOG fallback) receive strong Gaussian blurring of the head region plus lighter full-box blurring. If no person localization is available, the module *fails closed* by blurring the entire frame. Raw frames are never written to disk.

6.5 Agent Implementations

6.5.1 CameraAgent

The camera loop reads frames, strides to the target inference rate, and calls `model.track(persist=True, tracker="bytetrack.yaml")`, yielding boxes with stable track identifiers in one call. The `ZoneRuleEngine` maintains dwell-time dictionaries keyed by (track, zone) and static-object records keyed by track, firing each rule at most once per key. The optional `ClipAnomalyScorer` encodes three “normal”

and five “anomalous” text prompts once at startup; per sampled frame, the anomaly score is the softmax mass on anomalous prompts. It is retained as an experimental component, not a validated detector: its calibration AUC is 0.308 (Chapter 7). For coordination, the agent subscribes to task announcements, computes its bid utility, and on award performs verification by re-running detection at `imgsz=960`, `conf=0.25` on its most recent frame, reporting the maximum person confidence found.

6.5.2 FusionAgent and CoordinatorAgent

The FusionAgent is callback-driven (one handler per incoming event) with a 1 Hz maintenance tick that flushes hypotheses whose window has closed; confidence follows Equation 5.1 exactly. The CoordinatorAgent implements the auction as: publish task, sleep the bidding window, select the maximal bid, publish award, and block (with timeout) on the verification topic—approximately 120 lines including the round-robin baseline and message accounting used by the evaluation.

6.5.3 PolicyAgent

The policy decision is a pure function of (severity map, thresholds, cooldown state, verification adjustment), executing in well under a millisecond; the measured decision latency in our runs averages 0.3 ms excluding verification rounds. Every code path terminates in an audit record.

6.5.4 ExplanationAgent

The explanation pipeline is implemented as four explicit methods (`collect`, `describe`, `draft`, `guardrail`) over a typed state object. The draft call requests `response_format=json_object` from the model; the guardrail validates schema completeness, then extracts all `ev_*` identifiers from both the citation list and the free text via regular expression and checks set inclusion against the hypothesis’s actual event identifiers. The unit test suite includes a hallucination probe verifying that a draft citing a fabricated identifier is rejected and the template fallback engages.

6.6 Scenario Replay and Baselines

`scenarios/replay.py` instantiates the full system from a JSON manifest declaring sensors, zone polygons, and ground-truth events, then runs one of four modes: `mas-auction`, `mas-rules` (round-robin), `mas-nocoord` (coordination disabled), or `centralized`. The centralized baseline processes sensor sources *sequentially in a single process*. The modes do not isolate architecture perfectly: `replay.py` paces MAS modes but not the centralized mode, while fusion windows and cooldowns are measured in wall-clock time. Time-to-alert comparisons must therefore be interpreted as confounded rather than as a pure concurrency measurement. Every run serializes alerts, ground truth, and per-agent metrics to a result JSON.

6.7 Evaluation Tooling and Dashboard

`eval/metrics.py` matches alerts to ground-truth events by incident family and time tolerance (± 5 s), computing event-level precision/recall/F1, mean time-to-alert, false alerts per hour, and coordination overhead, and emits a summary CSV. The Streamlit console renders the alert feed with severity filtering, incident detail with the agentic report and anonymized evidence gallery, and acknowledge/dismiss buttons that write operator audit records. A separate Next.js console in `frontend/` provides an incident rail and wall-oriented view. Zone geometry and scenario-specific

thresholds live in `scenarios/*.json`; the previously documented `top-level configs/` directory does not exist.

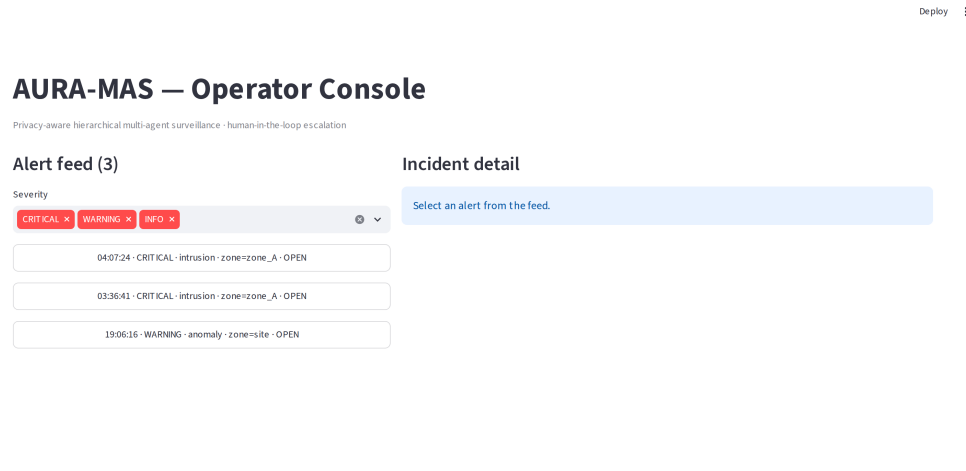


Figure 6.1: Live Streamlit operator-console capture reading local alert JSONL files. The view shows three open alerts: two **CRITICAL** intrusion alerts in **zone_A** and one **WARNING** anomaly alert. This is a live interface capture, not a 373-run MQTT/Redis measurement.

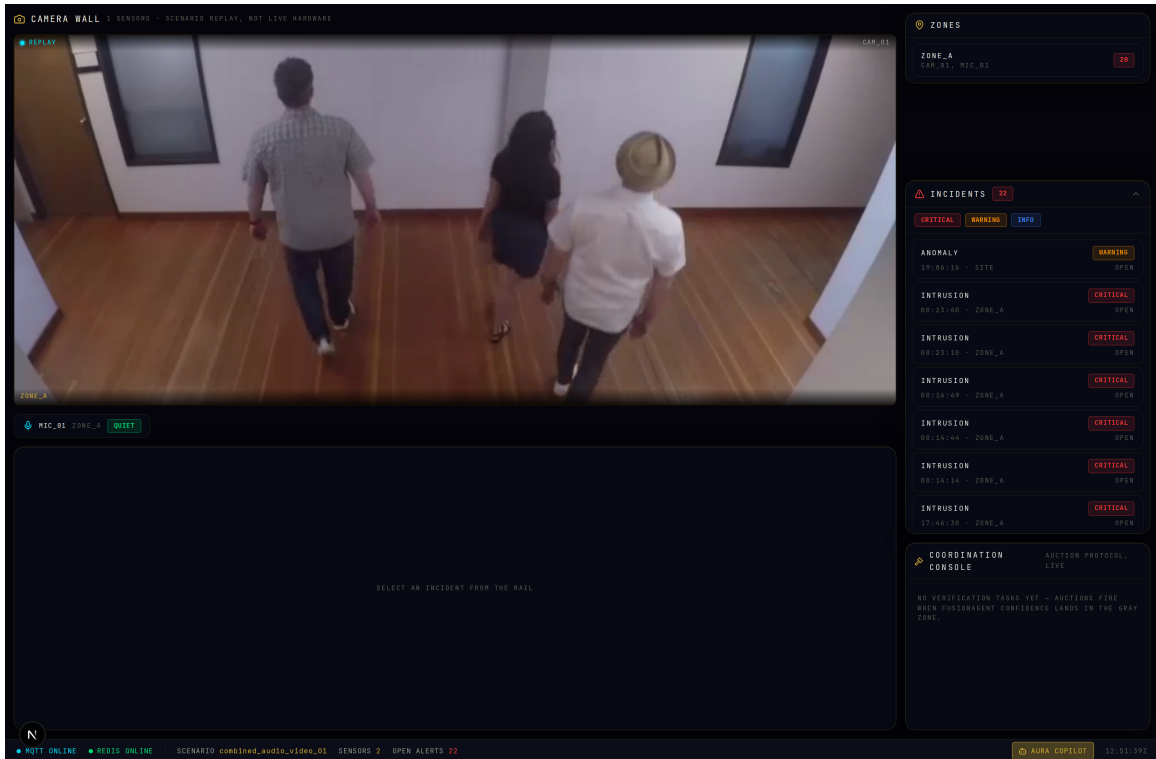


Figure 6.2: Live Next.js operator-console capture for the `combined_audio_video_01` replay. The status indicators describe the running local environment and must not be read as evidence that MQTT or Redis was exercised in the campaign, which used LocalBus.

6.8 Testing

The repository contains 26 offline tests. The model-free subset validates wildcard bus routing, monotone noisy-OR behaviour, auction best-bidder selection, policy thresholds and cooldown suppression, metric computation, and the guardrail hallucination probe. This supports test-driven iteration on coordination logic independently of perception, but it does not substitute for campaign evidence about performance.

6.9 Conclusion

The implementation provides the components described in Chapter 5, together with fallbacks, single-process tests, and serialized run artifacts. The next chapter evaluates those artifacts.

Part V

Evaluation and Results

Chapter 7

Evaluation and Results

7.1 Introduction

This chapter evaluates AURA-MAS as a system. It uses a controlled architectural ablation over nine replay scenarios to address RQ1–RQ5. The chapter sets out the experiment, the metrics, the statistical procedure, the results, and the limits of the evidence.

Revision notice

The first compiled version reported one run of `demo_site_01` and concluded that auction coordination improved precision and halved false alerts. The later 373-run campaign (`results/evaluation_campaign_v2_notes.md`) does not support that conclusion. Section 7.2.6 shows that no pairwise architecture difference survives correction for multiple comparisons. The coordination and multimodal-fusion claims are corrected in Sections 7.4 and 7.3.4; Appendix A records each replacement.

7.2 Experimental Setup

7.2.1 Evaluation Methodology

Component metrics such as mAP and tracking HOTA do not describe the whole alerting system. Operators encounter alert delay, false alerts, and the cost of coordination. Following ETISEO [52] and MLOps evaluation practice [53, 84], the experiment replays recorded sources through the complete agent stack and scores emitted alerts against a ground-truth manifest. Each variant receives the same recorded input. This controls the input, although Section 7.5.2 explains why it does not fully isolate architecture.

7.2.2 Scenario Corpus

Evaluation uses nine scenario manifests rather than the single demonstrator scenario of the first version. Each manifest declares its sensors, zone polygons, and ground-truth incident intervals; full listings appear in Appendix B. Table 7.1 summarizes the corpus.

Table 7.1: Scenario corpus. Sources are real, citably-licensed recordings (Appendix C); `loitering_01` is a deliberate true-negative probe with empty ground truth.

Scenario	Sensors	GT	Purpose / source
<code>intrusion_01</code>	2 cam, 1 mic	2	Zone intrusion + audio; pedestrian footage
<code>demo_site_01</code>	2 cam, 1 mic	3	Multi-zone demonstrator (§7.5.3)
<code>combined_audio_video_01</code>	1 cam, 1 mic	2	Cross-modal corroboration probe
<code>abandoned_object_01</code>	1 cam	1	Static-object rule; ABODA
<code>fight_01</code>	1 cam	1	CLIP semantic anomaly; AIRTLab [85]
<code>audio_glass_break_01</code>	1 mic	1	Audio-only, <i>security</i> family; ESC-50 [86]
<code>audio_alarm_siren_01</code>	1 mic	1	Audio-only, <i>hazard</i> family; ESC-50
<code>audio_alarm_clock_01</code>	1 mic	1	Audio class discrimination; ESC-50
<code>loitering_01</code>	1 cam	0	True-negative probe; false-alert floor

The corpus combines three audio-only scenarios, five video-only or video-dominant scenarios, and one cross-modal scenario. Aggregate means therefore mix settings in which different modalities carry the signal. The corpus also has only nine scenarios. Section 7.3.2 shows what that sample size can support.

7.2.3 Architecture Variants

Four variants are compared:

- **Centralized:** a single sequential process iterating over sensors—the classical VMS analytics topology (baseline for RQ1);
- **MAS-nocoord:** concurrent agents, fusion and policy active, coordination disabled (baseline for RQ2);
- **MAS-rules:** coordination via round-robin task assignment (non-market baseline for RQ2);
- **MAS-auction:** the full AURA-MAS design with sealed-bid verification auctions.

The first version described the variants as differing only in architecture. That is not accurate. The centralized variant runs sequentially and without pacing, which changes the effective duration of the wall-clock fusion window and alert cooldown. Section 7.5.2 discusses this pacing confound.

Runs were executed on a 4-vCPU CPU-only sandbox (YOLO11n at 5 FPS per camera).

7.2.4 Campaign Structure

The campaign crosses scenario, mode, modality configuration, audio backend, and repetition. `aura_mas/scripts/run_campaign.py` starts a fresh subprocess for each run (Figure 7.1) because model, tracker, and thread-pool state do not reset reliably in one process. The campaign completed 373 runs without a failure. The main analysis uses the 180 audio-visual YAMNet runs with $N = 5$ repetitions.

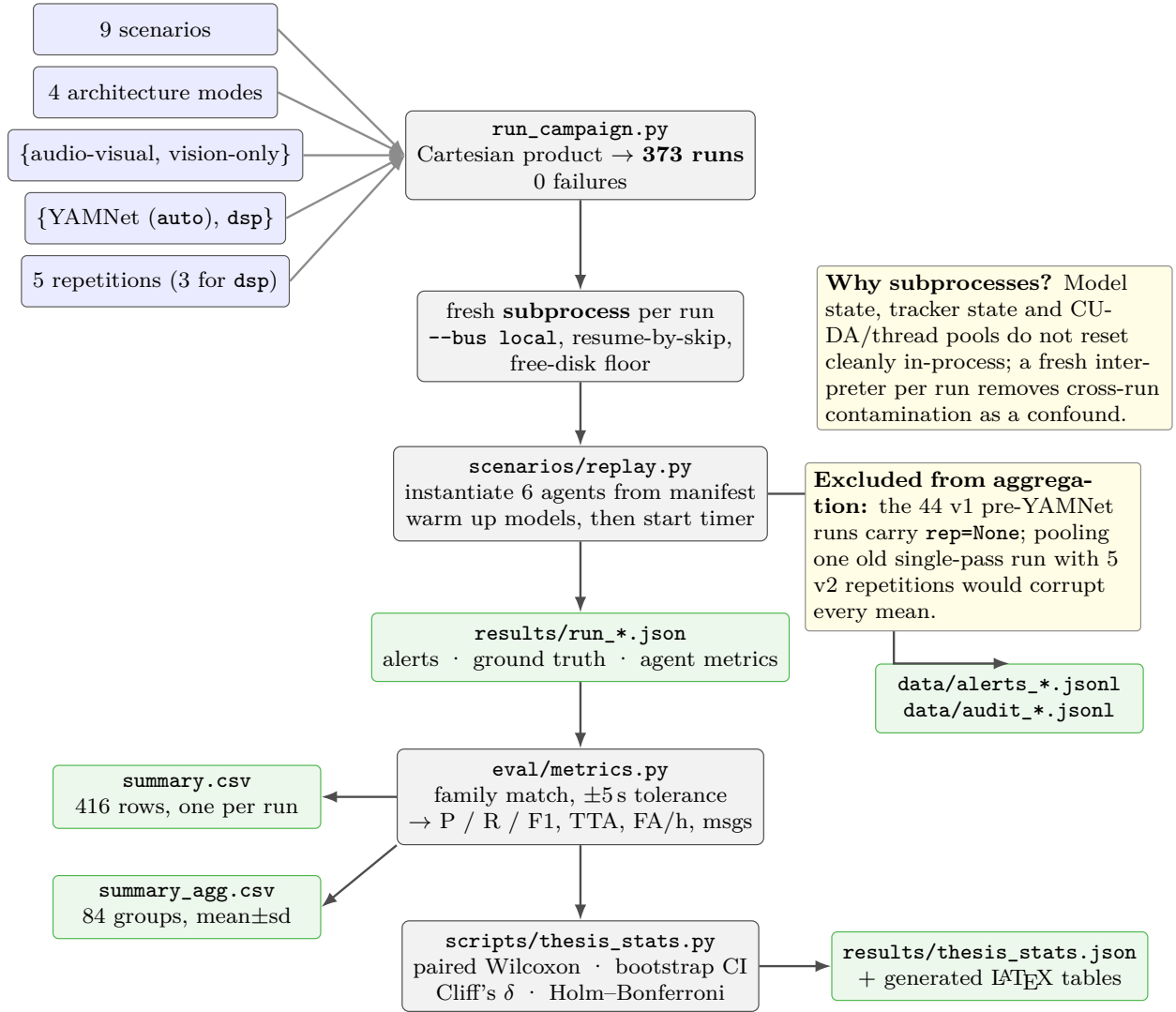


Figure 7.1: Evaluation campaign grid and scoring pipeline. Contribution C5 is this apparatus, not any particular number it produced.

7.2.5 Metrics

An alert matches a ground-truth incident if it carries the same incident *family* (Appendix D) and falls within a ± 5 s tolerance of the incident interval. We report event-level **precision**, **recall**, and **F1**; **mean time-to-alert** (TTA); **false alerts per hour** (FA/h); and **coordination overhead** (count of task/bid/award/verification messages). Section 7.5.2 documents three measurement limitations of this scheme—family-level matching, greedy assignment, and the FA/h denominator—each of which bounds what the corresponding number can mean.

7.2.6 Statistical Methodology

The first version of this chapter reported means and standard deviations over $N = 5$ repetitions with no interval, test, or effect size. That is insufficient here for a specific reason: on scenarios carrying one to three ground-truth events, per-run F1 takes few distinct values, so a reported “ 0.800 ± 0.447 ” describes an outcome that was 1.0 in four runs and 0.0 in one—a Bernoulli flip, not a dispersion estimate.

The campaign grid is a *paired* design: every (scenario, repetition) cell is observed under all four modes on identical inputs. The analysis therefore uses the paired difference as its unit and applies:

- **Wilcoxon signed-rank tests** on matched (scenario, repetition) pairs, rather than a paired t -test, because per-run F1 is discrete and bounded;
- **Holm–Bonferroni correction** across all six pairwise mode comparisons—quoting the smallest of six uncorrected p -values would be a multiplicity error;
- **Cliff’s δ** as a non-parametric effect size, reported alongside every p -value;
- **Bootstrap percentile confidence intervals** (10,000 resamples, fixed seed) on per-mode means and on paired differences.

All values are produced by `aura_mas/scripts/thesis_stats.py` into `results/thesis_stats.json` and emitted directly as L^AT_EX tables, so no number in this chapter is transcribed by hand. The analysis set comprises $n = 45$ fully-observed paired cells (9 scenarios $\times$ 5 repetitions).

7.3 Results

7.3.1 Architectural Ablation Across the Corpus

Table 7.2 reports event-level F1 for every scenario and mode; Figure 7.2 presents the same data as a heatmap.

Table 7.2: Event-level F1 per scenario and architecture (mean $\pm$ sd over $N = 5$ repetitions, YAMNet backend). Bold marks the best mode per scenario. [†]loitering_01 has empty ground truth by construction, so F1 is 0 for every mode.

Scenario	Centralized	MAS-nocoord	MAS-rules	MAS-auction
abandoned_object_01	0.667$\pm$0.000	0.534 $\pm$ 0.298	0.200 $\pm$ 0.447	0.200 $\pm$ 0.447
audio_alarm_clock_01	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000
audio_alarm_siren_01	0.200 $\pm$ 0.447	0.600 $\pm$ 0.548	0.800$\pm$0.447	0.800$\pm$0.447
audio_glass_break_01	0.000 $\pm$ 0.000	0.800$\pm$0.447	0.800$\pm$0.447	0.800$\pm$0.447
combined_audio_video_01	0.467$\pm$0.274	0.400 $\pm$ 0.224	0.200 $\pm$ 0.274	0.100 $\pm$ 0.224
demo_site_01	0.773 $\pm$ 0.059	0.781$\pm$0.204	0.629 $\pm$ 0.053	0.667 $\pm$ 0.000
fight_01	1.000$\pm$0.000	0.800 $\pm$ 0.447	0.400 $\pm$ 0.548	0.200 $\pm$ 0.447
intrusion_01	0.567$\pm$0.091	0.280 $\pm$ 0.259	0.380 $\pm$ 0.217	0.300 $\pm$ 0.274
loitering_01 [†]	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000

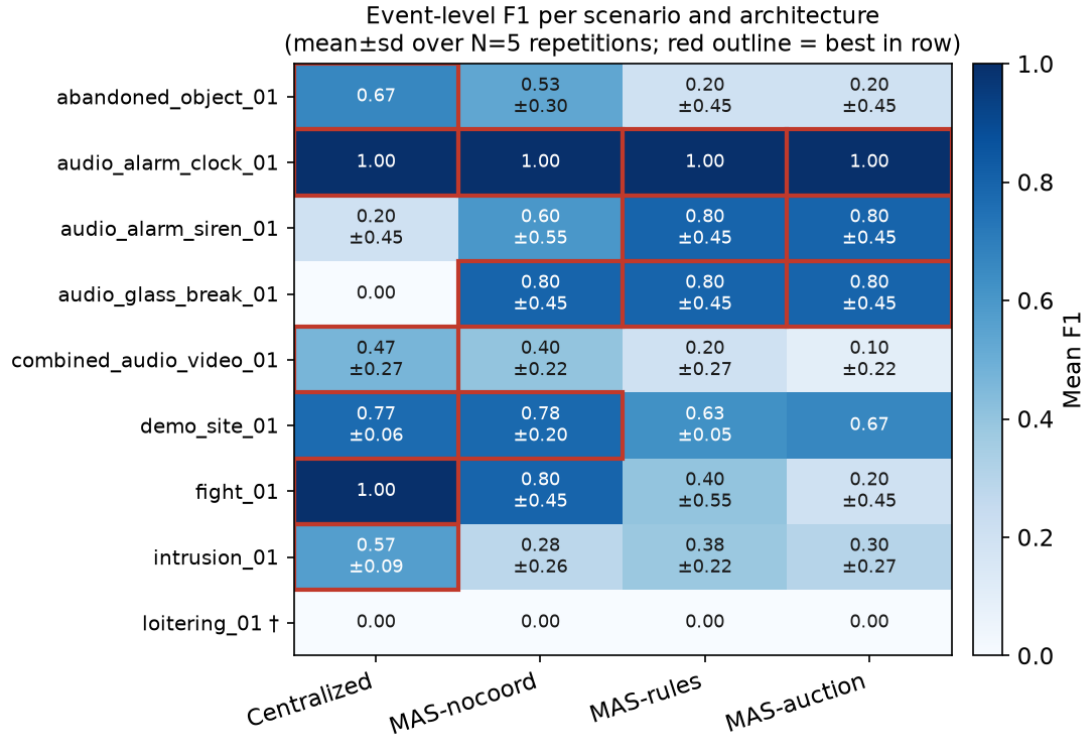


Figure 7.2: Event-level F1 across the nine-scenario corpus. No mode dominates: the centralized baseline wins on the video-dominant scenarios, while all three MAS variants win on the audio-only scenarios.

The per-scenario results depend on the modality carrying the signal. The centralized baseline is best on `fight_01`, `abandoned_object_01`, `intrusion_01`, and `combined_audio_video_01`, which are video-dominant scenarios. All three MAS variants beat it on `audio_glass_break_01` (0.800 vs. 0.000) and `audio_alarm_siren_01`. The overall mean is therefore strongly affected by the corpus’s modality mix.

With that caveat stated, the marginal means are given in Table 7.3 and Figure 7.3.

Table 7.3: Marginal mean F1 per mode with bootstrap confidence intervals ($n = 45$ paired observations per mode).

Mode	n	Mean	Median	95% CI (bootstrap)
Centralized	45	0.519	0.667	[0.401 , 0.633]
MAS-nocoord	45	0.577	0.667	[0.455 , 0.697]
MAS-rules	45	0.490	0.500	[0.361 , 0.620]
MAS-auction	45	0.452	0.500	[0.322 , 0.589]

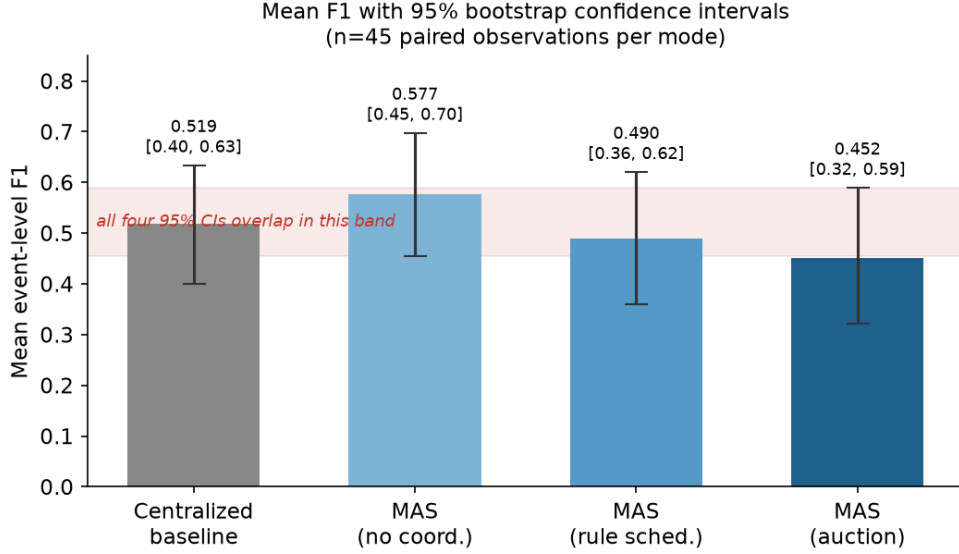


Figure 7.3: Mean event-level F1 with 95% bootstrap confidence intervals. All four intervals overlap in the shaded band.

On point estimates, the ordering is MAS-nocoord (0.577) > Centralized (0.519) > MAS-rules (0.490) > MAS-auction (0.452). **The full AURA-MAS configuration has the lowest mean F1 of the four variants on this corpus.** This directly contradicts the first version’s conclusion and is reported here as the primary result.

7.3.2 Inferential Analysis

Point-estimate orderings over 45 observations invite over-reading, so Table 7.4 gives the paired tests and Figure 7.4 the corresponding intervals.

Table 7.4: Pairwise architecture comparisons on F1, paired on (scenario, repetition). Δ mean is mode A minus mode B; δ is Cliff’s delta; p_{Holm} is Holm–Bonferroni-adjusted across all six comparisons.

Mode A	Mode B	Δ mean	95% CI	p	p_{Holm}	δ
MAS-nocoord	MAS-auction	+0.125	[0.022, 0.229]	0.024	0.145	+0.146
MAS-nocoord	MAS-rules	+0.087	[-0.015, 0.189]	0.097	0.487	+0.107
Centralized	MAS-nocoord	-0.058	[-0.205, 0.082]	0.432	1.000	-0.085
MAS-rules	MAS-auction	+0.038	[-0.066, 0.144]	0.552	1.000	+0.044
Centralized	MAS-auction	+0.067	[-0.113, 0.239]	0.562	1.000	+0.067
Centralized	MAS-rules	+0.029	[-0.136, 0.189]	0.671	1.000	+0.028

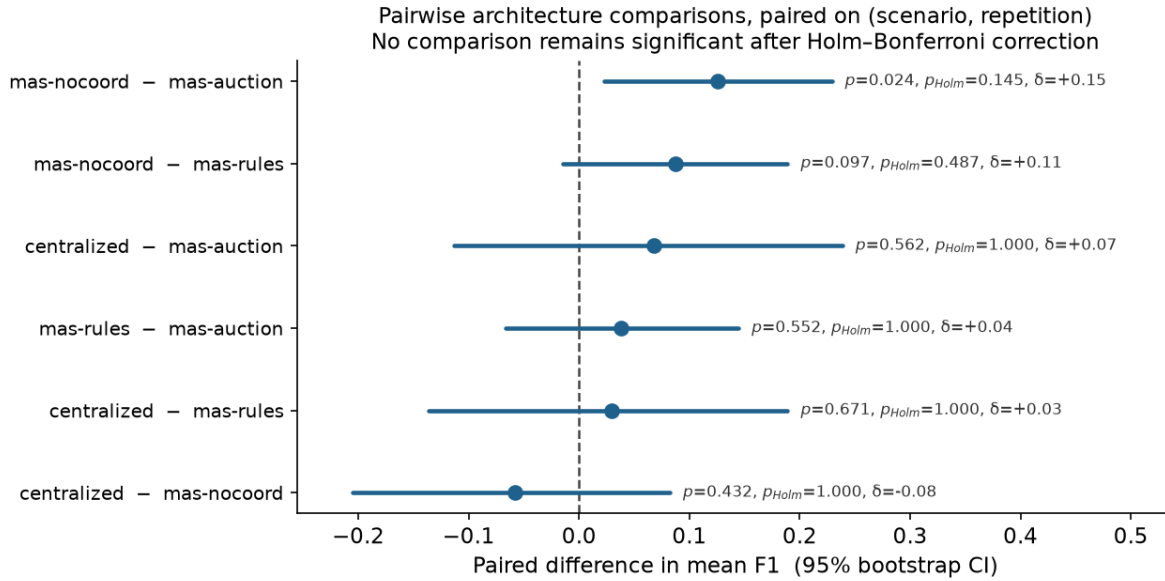


Figure 7.4: Paired mean differences in F1 with 95% bootstrap confidence intervals. Every interval either contains zero or, after multiplicity correction, cannot be distinguished from it.

Principal empirical finding

Across the nine-scenario corpus, **none of the six pairwise architecture comparisons on F1 is statistically significant after Holm–Bonferroni correction**. The largest raw effect—MAS-nocoord over MAS-auction, $\Delta = +0.125$, $p = 0.024$ —becomes $p_{Holm} = 0.145$, with Cliff’s $\delta = +0.146$, a negligible effect size. All four bootstrap confidence intervals overlap substantially. This campaign cannot distinguish the four architectures on detection quality. The point-estimate ordering should not be read as a performance ranking.

For the observed MAS-nocoord/MAS-auction difference, the standardized paired effect size is $d_z = 0.35$. Detecting an effect of that size with 80% power at $\alpha = 0.05$ requires about **65 paired observations**; this campaign has 45 (Figure 7.5). At five repetitions per scenario, that corresponds to about **13 scenarios**, not nine. A larger corpus is needed before treating the ordering as evidence of an architecture effect.

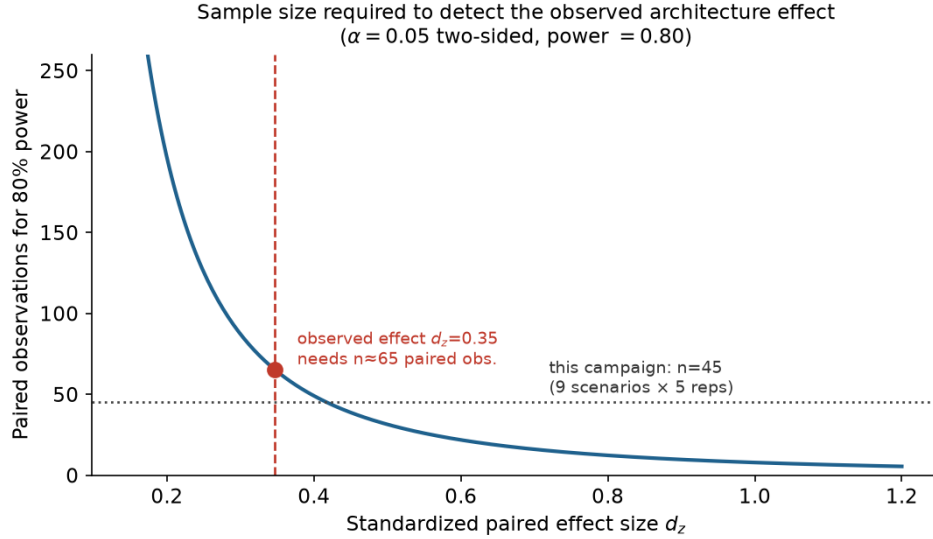


Figure 7.5: Paired observations required for 80% power as a function of standardized effect size, with this campaign’s position marked.

A sensitivity check excluding the empty-ground-truth `loitering_01` probe leaves the conclusion unchanged: the means rise (nocoord 0.649, centralized 0.584, rules 0.551, auction 0.508) but no comparison becomes significant after correction (`results/thesis_stats.json`, key `sensitivity_excluding_probe`).

7.3.3 Audio Backend Ablation: DSP Fallback vs. YAMNet

The one manipulation in this campaign with an unambiguous effect is the audio backend. Replaying the three audio-only scenarios under both backends ($N = 3$ per cell), the DSP fallback scores $F1 = 0.000 \pm 0.000$ in *every* repetition, across all four coordination modes and all three scenarios, while YAMNet reaches 0.6–1.0 on most scenario/mode pairs and a clean 1.000 ± 0.000 on `audio_alarm_clock_01` (Figure 7.6).

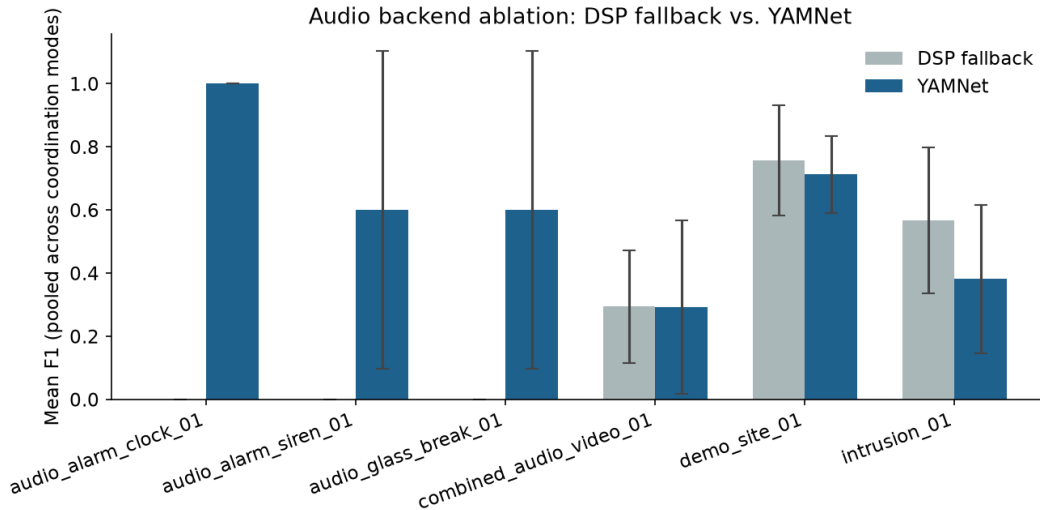


Figure 7.6: F1 by audio backend across the three audio-only scenarios.

This is a matching result rather than a direct comparison of audio detection quality. The DSP path emits the generic `audio_anomaly` type, mapped to the *violence_or_hazard* family. The audio-only manifests use *security* and *hazard* ground-truth families. Family-level matching therefore

cannot count a DSP detection as a true positive, even if it detects the transient. The 0.000 result shows a label-space mismatch. Under this protocol, class-specific audio typing is required for the modality to score as usable.

7.3.4 Analysis by Research Question

RQ1 (Architecture). On `demo_site_01`, mean TTA falls from 15.07 ± 3.11 s (centralized) to 2.96 ± 0.14 s (MAS-auction), an 80.4% reduction. That result does not extend to the wider corpus. Across the seven scenarios in which all four modes produce a matched alert, the paired means are centralized 9.47 s, MAS-nocoord 8.87 s, MAS-rules 6.60 s, and MAS-auction 8.46 s. The centralized-to-auction difference is 10.7%, with $p = 0.59$, and no pairwise TTA comparison is significant after correction. `demo_site_01` is the only scenario in which the centralized variant is much slower. Its own $N = 5$ result has $p = 0.0625$.

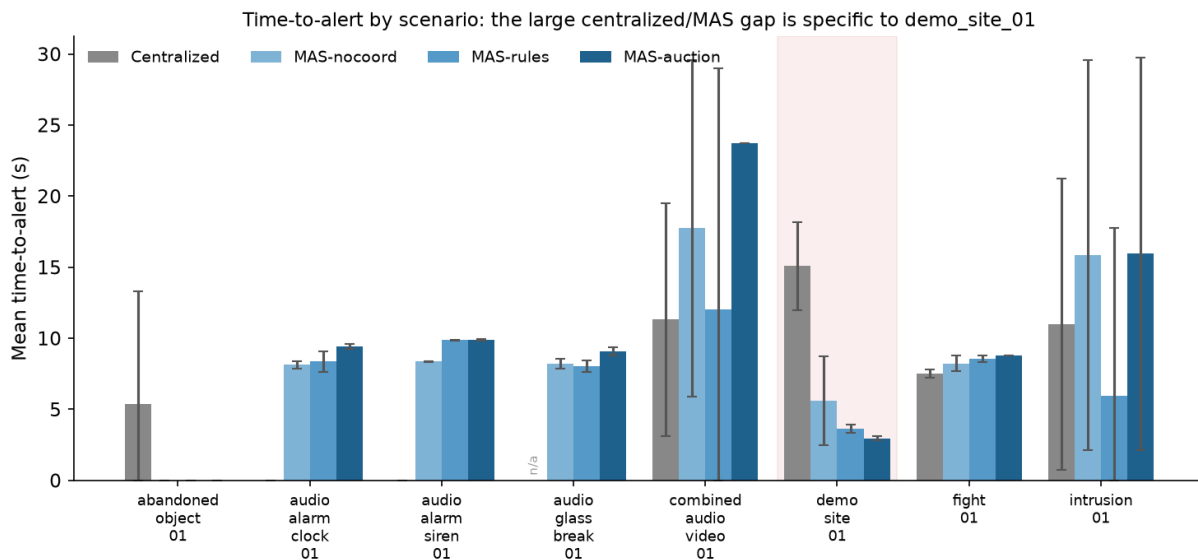


Figure 7.7: Mean time-to-alert by scenario. The large centralized/MAS gap is specific to `demo_site_01` (shaded); elsewhere the ordering flattens or reverses. “n/a” marks mode/scenario pairs that produced no matched alert.

Sequential processing can create head-of-line blocking, so a concurrency effect is plausible. Here, however, the result is confounded with pacing and appears clearly in only the scenario with the most sensors. RQ1 remains unresolved.

RQ2 (Coordination). Addressed in Section 7.4.

RQ3 (Multimodality). Audio zone tagging and event-family taxonomy unification allow audio and video events to enter the same hypothesis (Chapter 6). Four campaign alerts list both a camera and a microphone in their `sensors` field. The mechanism therefore runs end to end.

Revision notice

The first version of this chapter stated that on `combined_audio_video_01` “the resulting alert’s confidence (0.78) reflects the audio+video corroboration bonus of Equation 5.1.” **This is incorrect.** Direct inspection of every alert in the campaign (`results/cross_modal_fusion_audit.md`) shows that the 0.78 alert carries `sensors: ["cam_01"]` and exactly one fused event, so neither bonus term—both gated on $|M(H)| > 1$ and $|S(H)| > 1$ —can apply. The value 0.78 is precisely one unaided video event: $1 - (1 - 0.9 \times 0.867) = 0.780$. All 37 alerts at confidence 0.78 in the campaign are single-sensor.

Only 4 of 240 alerts are genuinely cross-modal because audio and video events must share a (family, zone) key within the same 6 s window. Their effect on reported confidence cannot be measured because noisy-OR reaches the clamp first. In the example in Figure 7.8, four same-camera detections already produce 0.9992. The two +0.05 bonuses are then absorbed by the $[0, 1]$ clamp, and a video-only hypothesis reaches the same reported 1.0. All four cross-modal alerts report 1.0.

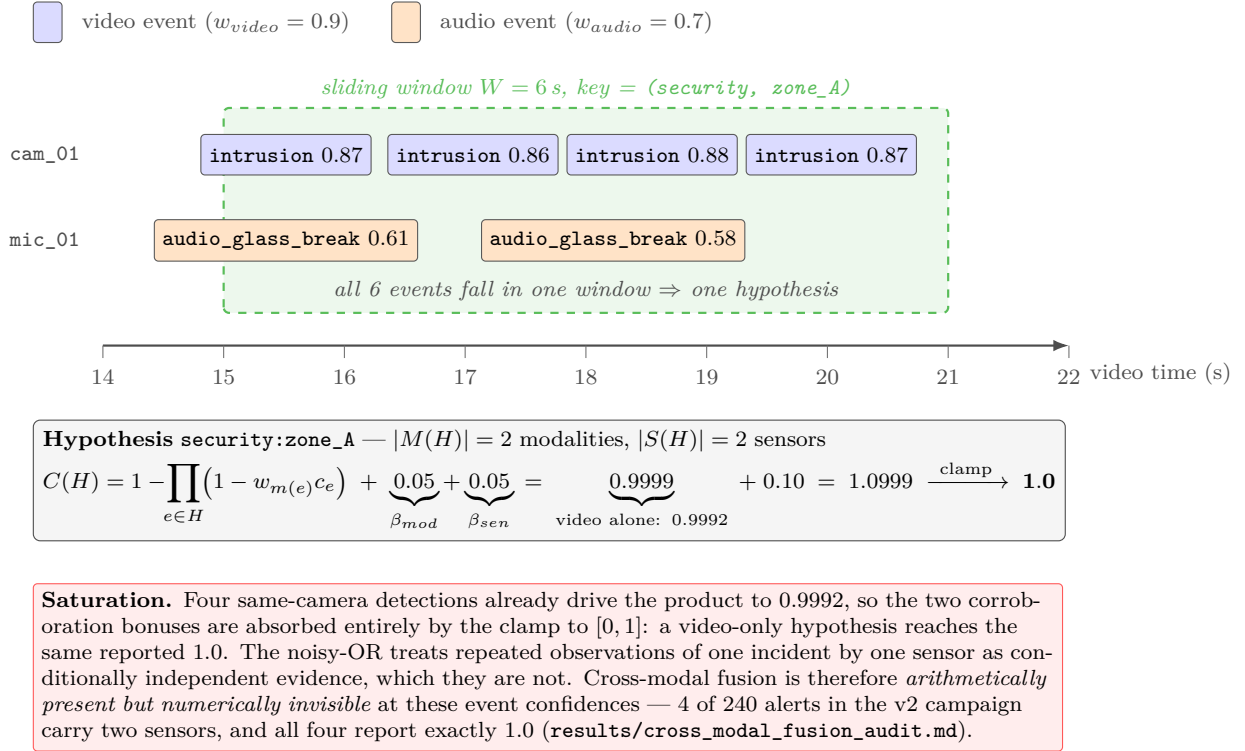


Figure 7.8: Hypothesis formation and noisy-OR saturation on combined_audio_video_01. Confidences are the values recorded in the run artifacts.

Noisy-OR treats repeated observations of one incident from one sensor as conditionally independent, although they are not. C3’s monotonicity guarantee still holds, but the score has already saturated at these event confidences. A later version should pool evidence per sensor or use a dependence-aware combination rule.

Separately, `Hypothesis.dominant_type()` labels all four cross-modal alerts `intrusion` rather than `audio_glass_break`, because video confidence always outscores audio confidence within the merged hypothesis. A reader of the raw alert log therefore sees no indication that audio contributed—a limitation of single-winner event-type labelling that motivates the `contributing_types` field discussed in Chapter 8.

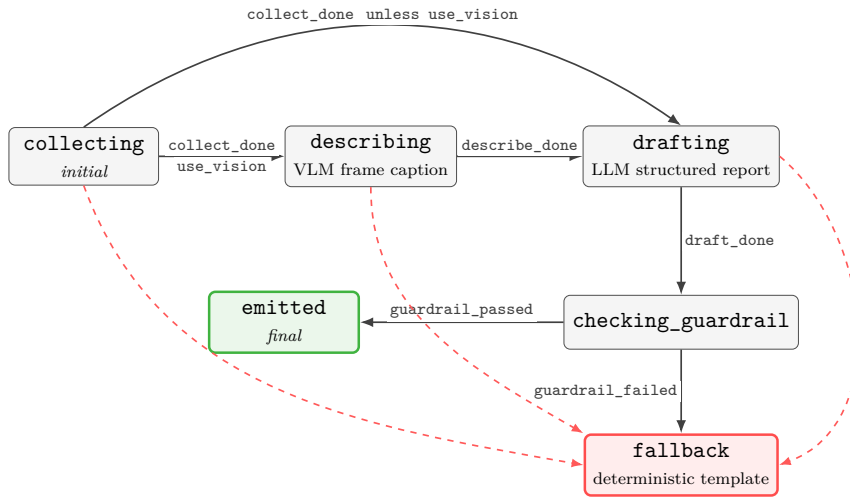
RQ4 (Agentic explanation). Explanation generation occurs strictly after alert emission by construction, so it cannot affect alerting latency; measured policy decision latency averages 0.3 ms.

Revision notice

The first version stated that “in replay runs, all accepted reports cited only identifiers present in the event log, i.e., zero uncaught hallucinated citations.” This claim is **vacuous as stated**: `run_campaign.py` never passes `--llm`, so across all 373 runs the LLM path never executed and every `explanation` field written to disk is the deterministic fallback template (results/explanation_judge_notes.md). No report was generated by a model, so none could have been found grounded.

C4 has structural evidence and one observed model run. The explanation pipeline is a finite state machine (Figure 7.9); every path to `emitted` passes the guardrail, and every failure uses the

deterministic template. Inspection and unit tests can verify that control flow without making claims about a model. In one post-campaign run, a local `qwen2.5-vl:3b` report failed the guardrail and triggered the template (`aura.guardrail.passed: false, aura.explanation.fallback_triggered: true` in `data/otel_spans.jsonl`). This single result shows that the mechanism can fire, but it does not evaluate the guardrail or the local model. C4 remains a structural design contribution.



Dashed red: the fail transition, available from every non-final state. No path reaches `emitted` except through a passing guardrail check — the structural property behind contribution C4. The guardrail rejects any report citing an `evidence_id` absent from the alert; a structurally invalid response is not retried, it routes straight to fallback.

Figure 7.9: ExplanationAgent state machine (`aura_mas/agents/explanation_fsm.py`). The guardrail is the sole gate on the `emitted` state.

RQ5 (Governance). This question concerns invariants rather than effect sizes. Every exported crop passed through the anonymization choke point in `aura_mas/core/privacy.py`. The campaign wrote 1,737 anonymized evidence files, with examples in Figure 7.10, and persisted or transmitted no raw frame. Each alert and suppression produced an audit record across all 373 runs, with a separate audit JSONL file for each run.



Figure 7.10: Anonymized alert evidence written by the privacy choke point during the campaign. Person regions receive graded Gaussian blurring, strongest over the head region; if no person localization is available the module fails closed and blurs the whole frame. Filenames are retained so any panel can be traced to its run.

Two boundaries on this claim should be stated plainly. First, anonymization *efficacy* is asserted structurally—every export traverses the choke point—but never measured; no re-identification attack was attempted against the blurred crops. Second, the audit stream is append-only by convention, not cryptographically immutable; the word “immutable” in the first version overstates it, and Chapter 8 lists hash-chaining as the concrete remedy.

7.4 Coordination: Retraction and Reassessment

RQ2 asks whether auction-based verification allocation improves alert precision over uncoordinated alerting, and at what communication cost. This section answers it, and in doing so retracts the first version’s central claim for contribution C2.

7.4.1 What the first version claimed

The first version reported, from a single run, that auction coordination halved false alerts and restored precision, and concluded that because MAS-rules and MAS-auction pay a similar message cost, “the difference is attributable to *who* is asked to verify, not merely that verification happens—validating the utility-bid design in which field-of-view overlap and origin-independence, not an arbitrary rotation, dominate the bid.”

7.4.2 Why that claim cannot be supported

Revision notice

The utility-bid claim is **retracted**. The bid function is $u = b \cdot \kappa \cdot o$, where o is intended to be the camera's field-of-view overlap with the event zone. In the evaluated code, o is read as `task.get("fov_overlap", {}).get(self.agent_id, 0.5)` (aura_mas/agents/camera_agent.py:448), and the coordinator never populated `fov_overlap` when constructing the task. **Throughout all 373 runs, o was the constant 0.5 for every camera.**

The bid therefore reduced to $b \cdot \kappa$: 0.3 for the originating sensor and 1.0 for any other idle camera. On a two-camera site, it deterministically selects the other camera, as round-robin does. The measured code never activates the mechanism needed to test the original claim.

Figure 7.11 traces the protocol and marks where the inert term enters. This also explains mechanistically why MAS-auction and MAS-rules produce statistically indistinguishable results ($\Delta = +0.038$, $p_{\text{Holm}} = 1.000$, $\delta = +0.044$): under an inert overlap term on a two-camera site, they are close to the same allocator.

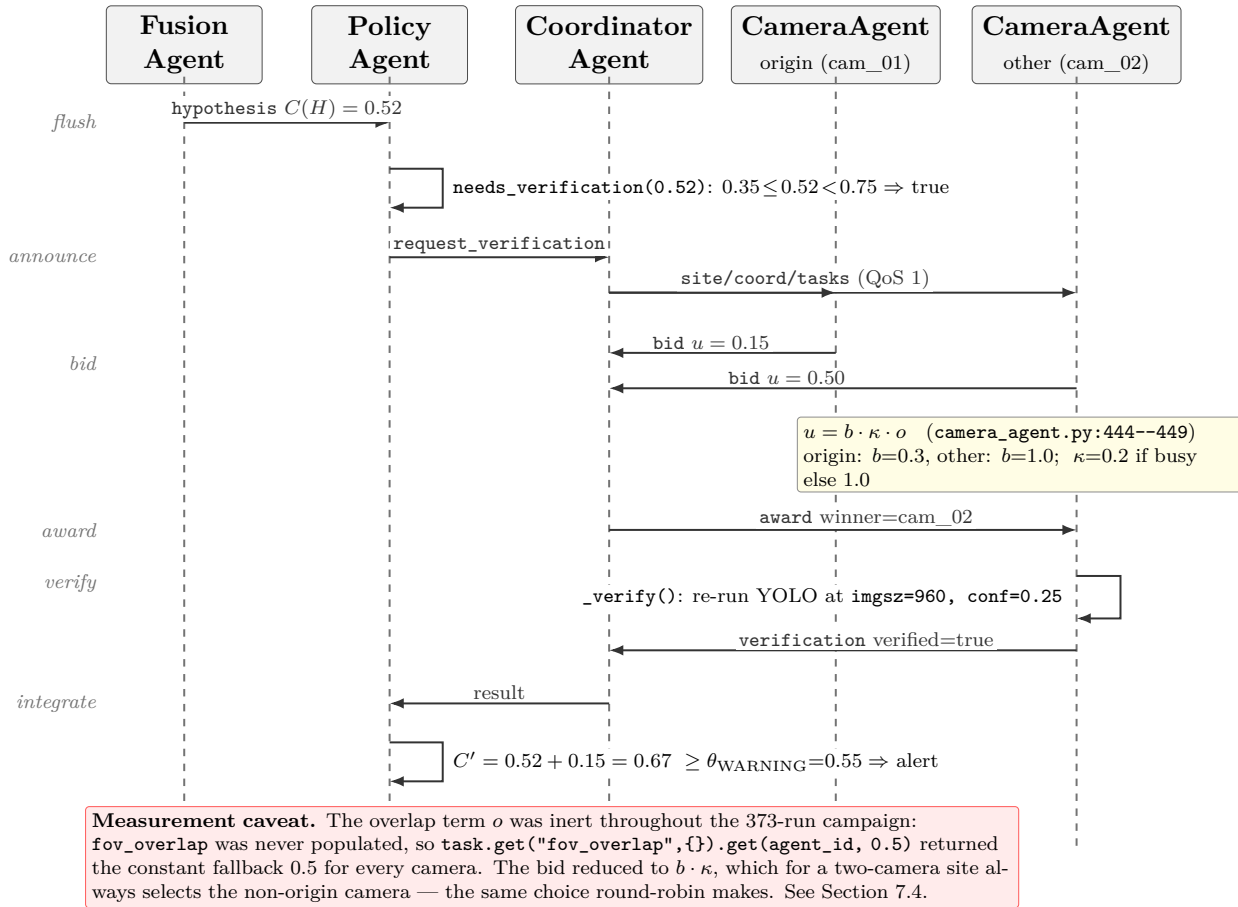


Figure 7.11: Auction-based active verification message sequence, annotated with the measurement caveat that the overlap term was inert during evaluation.

A transport path for `fov_overlap` has since been added to the coordinator and replay driver, but no scenario manifest populates the map, so the term remains at its fallback value and no result in this thesis exercises it. Implementing per-scenario overlap—by homography-based zone/frustum intersection, as sketched in Chapter 8—is the prerequisite for testing C2 at all.

7.4.3 The stability argument, and why it is also not supported

One possible reading is that auction coordination buys predictability rather than accuracy. On `demo_site_01`, MAS-auction precision is 0.667 ± 0.000 across five repetitions, while MAS-nocoord precision is 0.767 ± 0.224 . That reading would treat a lower mean with lower variance as useful to an operator.

We tested this rather than asserting it. Table 7.5 reports the per-mode precision variances and formal homogeneity tests.

Table 7.5: Precision dispersion by mode on `demo_site_01` ($N = 5$).

Mode	n	Mean precision	Variance
Centralized	5	0.933	0.02218
MAS-nocoord	5	0.767	0.04997
MAS-rules	5	0.600	0.00837
MAS-auction	5	0.667	0.00000

Levene’s test gives $W = 1.62$, $p = 0.224$; the Fligner–Killeen test gives $\chi^2 = 4.46$, $p = 0.216$. Pooled across the corpus, Levene gives $p = 0.515$. The variance difference is not statistically significant. MAS-auction’s 0.667 ± 0.000 also consists of five identical values from one scenario, not evidence of lower variance across operating conditions.

7.4.4 What can be said

The coordination overhead itself is measured cleanly: 5 ± 0.0 messages per run for MAS-auction versus 2 ± 0.0 for MAS-rules and 0 for MAS-nocoord (Figure 7.12). Even here a caveat applies: because every campaign run used the in-process `LocalBus`, these are counts of synchronous Python function calls, not network messages. The MQTT and Redis transports were never exercised in any reported result, so contribution C1’s substrate claim is **unevaluated**, and the “communication cost” in RQ2 is not a communication cost in the sense the research question intends.

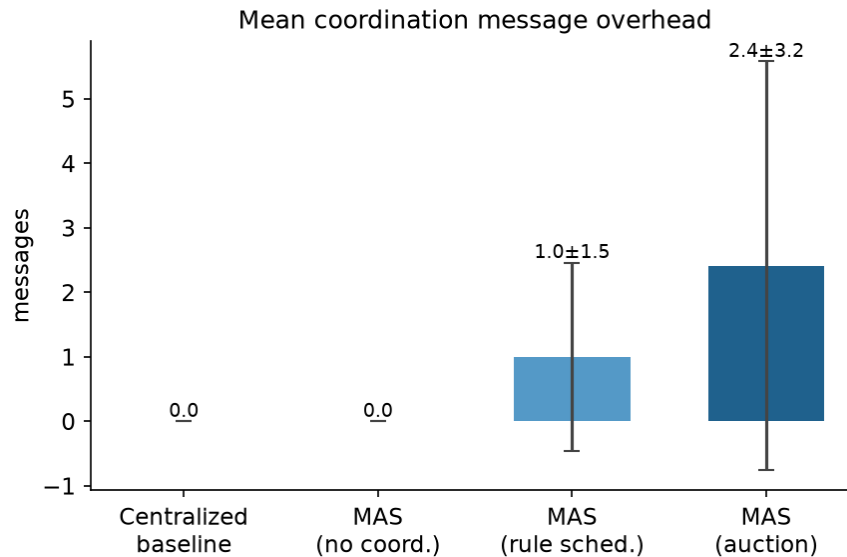


Figure 7.12: Coordination message overhead per run, counted as in-process bus publications.

RQ2 remains unresolved. The protocol is implemented, terminates, and has a bounded, measured message count. The experiment does not test whether utility-based allocation outperforms arbitrary allocation because the utility function’s informative term was inactive.

7.5 Discussion

7.5.1 Interpretation

The first version attributed responsiveness to concurrency and stability to market-based coordination. The corpus-wide data do not support either claim. The latency result is confined to one scenario and is not significant; the stability result fails its homogeneity test.

The evaluation supports three narrower observations:

1. **Architecture choice is not resolved as the main source of detection quality.** The four coordination topologies have statistically indistinguishable F1 over nine scenarios. Variation between scenarios is larger than variation between modes (Figure 7.2).
2. **Representation choices affected the measured score.** The audio-backend result comes from the label space, not the coordination topology: an output type that cannot match a ground-truth family scores zero. The cross-modal confidence result is similarly limited by saturation in the confidence representation.
3. **Several mechanisms were inactive during measurement.** The overlap term in the bid, the MQTT/Redis substrate, and the LLM explanation path were not exercised in the reported runs. Comparing code with artifacts exposed those gaps.

The evaluation apparatus is the clearest contribution of this thesis. Fresh-subprocess isolation, repetition indices, the preserved pre-fix baseline CSV, the methodology-change log in Appendix E, and the inferential analysis exposed results that a single favourable run did not show.

7.5.2 Threats to Validity and Limitations

Pacing confound (internal validity). `scenarios/replay.py` sets `realtime = mode != "centralized"`, so MAS modes pace playback to source frame rate while the centralized baseline runs unpaced and processes sensors sequentially to completion. Because `FusionAgent.window_seconds` and `PolicyAgent.cooldown_seconds` are wall-clock quantities, the effective temporal width of the fusion window and the alert cooldown differ between modes. The RQ1 latency comparison therefore measures pacing policy together with architecture, and is not attributable to architecture alone. A paced centralized variant is the required control and was not run.

Strawman baseline (construct validity). A production video-management system ingests streams concurrently at real time; it does not process them one after another to completion. The sequential baseline is weaker than the real comparator, which biases RQ1 in the MAS variants' favour by construction.

Family-level matching cannot distinguish event types. The matcher compares incident *families*, and the taxonomy assigns `intrusion`, `loitering`, `abandoned_object`, and `audio_glass_break` all to *security*. On `demo_site_01`, whose three ground-truth entries are all in that family, an `intrusion` alert can satisfy the `loitering` entry. The metric answers “did some security-relevant alert occur near this time,” not “was the right event detected.”

Tolerance width. Matching accepts $[t_{start} - 5, t_{end} + 5]$. For `demo_site_01`'s intrusion interval $[3, 35]$ on a 55 s scenario, the acceptance window spans roughly 73% of the run, which limits how much information precision, recall, and TTA can carry on that scenario.

Greedy assignment under-credits successful fusion. The matcher is a first-fit loop over ground truth in manifest order, not an optimal assignment. On `combined_audio_video_01`, whose manifest lists `intrusion` before `audio_glass_break` (both *security*), a single successfully-fused alert satisfies the first entry and leaves the second unmatched, capping measured recall below what the system achieved. A Hungarian assignment would remove the order dependence.

FA/h is degenerate at these run lengths. False alerts per hour is $fp/(wall\ seconds/3600)$. With runs as short as 9.1 s and false-positive counts in $\{0, 1, 2, 3\}$, the metric reaches 393.9 alert-s/hour from a single false positive. Figure 7.13 shows the measured values lying exactly on the $fp \times 3600/\text{duration}$ hyperbolae: the statistic reports run duration as much as alert quality, and should not be read as an operational false-alarm rate.

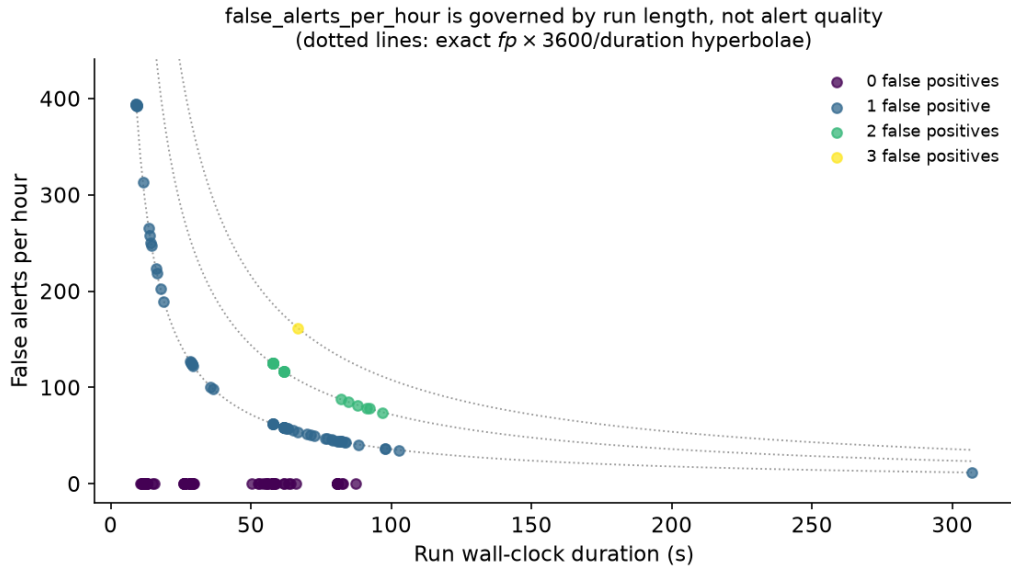


Figure 7.13: False alerts per hour against run duration. Every point lies on an exact hyperbola determined by its integer false-positive count, showing the metric is governed by the denominator.

Known-false ground truth on the demonstrator scenario. `scenarios/loitering_01.json` records an empirical check on the same clip and zone: the longest continuous single-track presence is 5.5 s, below the 8 s loitering threshold, so the loitering event `demo_site_01` declares at $t = 16\text{--}46$ s is not detectable by the implemented rule. That entry remains in the manifest and mechanically caps recall at $2/3$; 48 of 60 `demo_site_01` rows show exactly one false negative. The numbers in this chapter are reported with the annotation in place, and it should be regarded as a known defect of that scenario rather than a system limitation.

Verification examines the wrong frame. `CameraAgent._verify()` re-analyses `self._last_frame`, the most recent frame read, whereas the hypothesis under verification was formed from events up to 6 s earlier plus the bid window. Under real-time pacing the verified frame typically postdates the incident. The field is also written by the camera thread and read by the coordinator’s callback thread without synchronization. Additionally, `_verify()` counts only `person` detections regardless of hypothesis type, so verifying an `abandoned_object` or `audio_alarm` hypothesis asks a question unrelated to that hypothesis.

Semantic anomaly calibration. The CLIP zero-shot anomaly scorer measures Area Under the Curve (AUC) = 0.308 against held-out labelled frames—*worse than a random classifier*—root-caused to a prompt/scene domain mismatch: the normal prompts describe indoor warehouse scenes while much of the corpus is outdoor street footage. Because CLIP usage is held constant across all four variants, this does not bias the relative architectural comparison, but it means the semantic-anomaly channel contributed an anti-correlated signal, and no claim about CLIP’s standalone usefulness in this deployment is warranted.

Unjustified parameters. More than twenty constants govern the decision path—modality weights 0.9/0.7, bonuses 0.05, window 6 s, gray zone $[0.35, 0.75)$, thresholds 0.45/0.55/0.70, cooldown 20 s, verification $\pm 0.15/-0.20$, tolerance ± 5 s, dwell 8 s, static 10 s, IoU 0.6, and the per-class audio thresholds (Appendix D). None was tuned on a held-out split and only the CLIP anomaly

threshold was swept. No sensitivity analysis is offered, so the reported operating point should be treated as one arbitrary point in a large unexplored space.

No seed control. No random seed, thread count, or determinism flag is set anywhere. Run-to-run variation was observed and is handled by repetition rather than elimination, which inflates the variance against which all effects are measured and thus further reduces power.

Scale. Nine scenarios, at most three sensors, and runs of 8–73 s. Absolute values are demonstrator-scale; the corpus is too small to resolve the effects the research questions ask about, as Section 7.3.2 quantifies.

7.5.3 The Demonstrator Scenario as a Case Study

`demo_site_01` remains useful as an illustration of the full pipeline exercising every mechanism—multi-sensor concurrency, cross-modal fusion, gray-zone verification, thresholds, cooldowns, anonymization, and explanation—within a single 55 s replay. Table 7.6 reproduces its per-mode results for that purpose.

Table 7.6: Single-scenario results on `demo_site_01` ($N = 5$; mean $\pm$ sd). **Retained as an illustrative case study only.** These values are not representative of the corpus: see Table 7.2 and Section 7.3.2.

Mode	F1	Precision	Recall	TTA (s)	FA/h	Msgs
Centralized	0.773 $\pm$ 0.059	0.933 $\pm$ 0.149	0.667 $\pm$ 0.000	15.07 $\pm$ 3.11	7.4 $\pm$ 16.5	0
MAS-nocoord	0.781 $\pm$ 0.204	0.767 $\pm$ 0.224	0.800 $\pm$ 0.182	5.60 $\pm$ 3.14	29.8 $\pm$ 36.5	0
MAS-rules	0.629 $\pm$ 0.053	0.600 $\pm$ 0.091	0.667 $\pm$ 0.000	3.64 $\pm$ 0.27	61.4 $\pm$ 21.4	2 $\pm$ 0.0
MAS-auction	0.667 $\pm$ 0.000	0.667 $\pm$ 0.000	0.667 $\pm$ 0.000	2.96 $\pm$ 0.14	52.6 $\pm$ 5.9	5 $\pm$ 0.0

Read as a case study, this scenario is instructive precisely because it is misleading in isolation: it is the one scenario where the MAS latency advantage is large, the one where a known-false ground-truth entry caps recall, and the one whose three ground-truth events all collapse into a single incident family. It is a good demonstration and a poor experiment, and the distinction is the methodological lesson of this chapter.

Figures 7.14 and 7.15 present the corpus-pooled detection-quality and responsiveness views retained from the first version, now understood in light of the inferential analysis above.

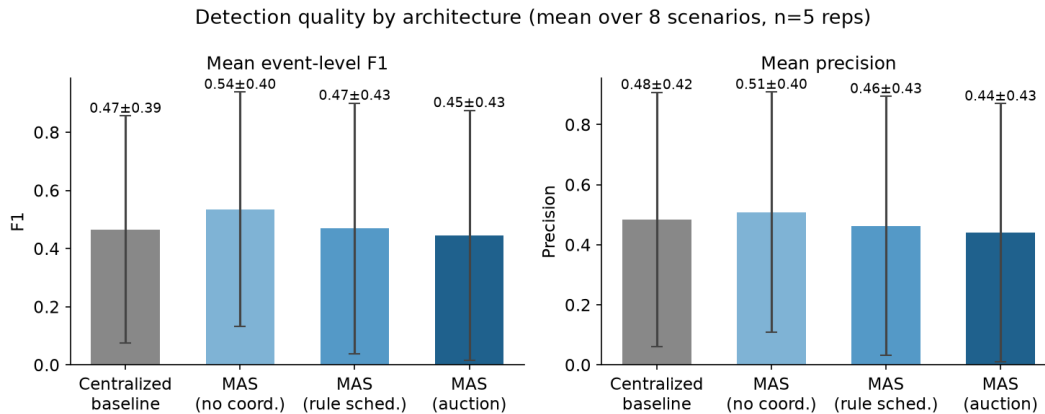


Figure 7.14: Detection quality pooled across scenarios and repetitions (error bars are pooled sample standard deviations, not confidence intervals; cf. Figure 7.3).

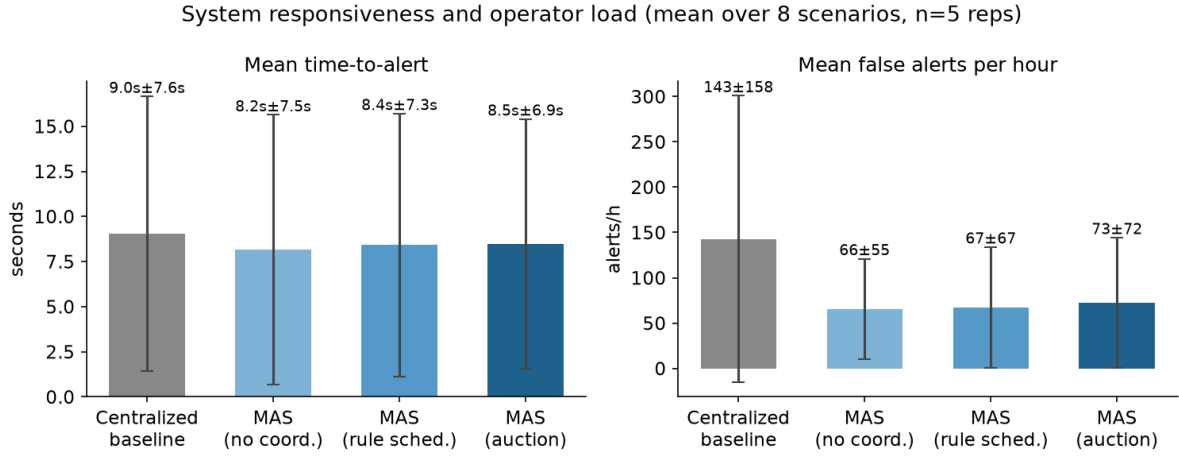


Figure 7.15: Pooled time-to-alert and false alerts per hour. The FA/h panel should be read with Figure 7.13 in mind.

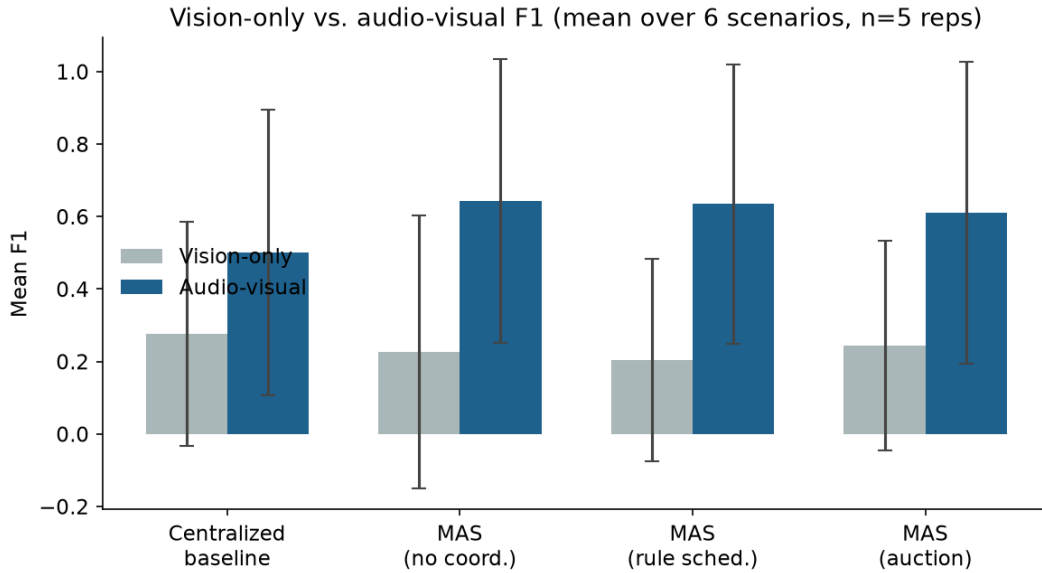


Figure 7.16: Vision-only versus audio-visual F1 across the audio-capable scenarios.

7.6 Conclusion

Across nine scenarios and 373 runs, the hierarchical MAS is not demonstrably faster to alert than the centralized pipeline once more than one scenario is considered. Pacing confounds RQ1. Auction coordination is not shown to improve precision or false-alert rate, and its central mechanism was inactive during measurement, so RQ2 remains unresolved. Cross-modal fusion works, but noisy-OR saturation makes its confidence contribution unmeasurable and cross-modal alerts are rare. The explanation guardrail is structurally supported and fired on the single real-model run, but the campaign did not exercise it. The privacy and audit pipeline held across all 373 runs, although anonymization efficacy was not measured.

Three research questions remain negative or unresolved. The replay artifacts, repeated runs, and inferential analysis made those limits visible. The final chapter revises the contribution claims and identifies the work needed to answer the open questions.

Part VI

Conclusion

Chapter 8

Conclusion and Future Work

8.1 Summary of the Work

AURA-MAS separates surveillance into a sequence of identifiable steps. Camera and audio agents emit structured events. The fusion agent groups related events into hypotheses, the coordinator may request verification, and the policy agent makes the alert decision. Explanations are generated only after that decision. This arrangement keeps raw video at the edge and prevents generative output from changing the alert path.

The final evaluation used 373 replay runs across nine scenario manifests. It also kept the pre-fix baseline and documented every change between the two measurement pipelines in Appendix E. The results do not support a performance ranking of the four architectures. After paired Wilcoxon tests and Holm–Bonferroni correction, none of the six pairwise F1 differences is statistically significant.

RQ1 remains unresolved. The 80.4% reduction in time-to-alert occurred only in `demo_site_01` ($n = 5$, $p = 0.0625$). Across the seven scenarios with all four modes, the reduction was 10.7% ($p = 0.59$). The replay driver also paces MAS modes but not the centralized mode, so the comparison mixes architecture with wall-clock behaviour of the fusion window and cooldown.

RQ2 cannot validate the auction as a spatial allocation method. The campaign manifests did not supply field-of-view overlap, so the bid’s informative term always used its 0.5 fallback. In a two-camera setting, that does not test whether the selected camera has the best viewpoint. The no-coordination and auction comparison is not significant after correction.

RQ3 confirms that cross-modal fusion can occur. Four of 240 campaign alerts include both a camera and a microphone. Its effect on confidence cannot be estimated from this corpus, however. The often-cited 0.78 alert was a single-camera video alert, and repeated events from one camera make the noisy-OR score reach one before an audio contribution can be seen.

RQ4 is supported as a design property, not as a campaign result. Unit tests and one post-campaign run show that the guardrail can reject an unsupported report. The campaign itself never enabled `--llm`, so all 373 explanations came from the deterministic template. RQ5 has the clearest evidence: the campaign wrote 1,737 anonymized crops and 373 audit files. The audit log is append-only, not cryptographically immutable, and the usefulness of the alerts for operators remains unmeasured.

8.2 Contributions Revisited

The final evaluation narrows the claims for the five contributions announced in Chapter 1.

C1 — Architecture. The system implements six agents and an operator console that separate perception, coordination, policy, explanation, and oversight. MQTT and Redis are available as transports, but the campaign used `LocalBus`, so it did not measure network cost.

- C2 — Active verification auction.** The task, bid, award, and verification flow is implemented and testable. Because campaign manifests left the spatial utility term unpopulated, the reported results do not show an advantage over round-robin allocation.
- C3 — Late fusion.** The noisy-OR implementation is monotone and accepts multi-sensor input. At the observed operating confidences it can saturate, and repeated events from one camera violate the model’s independence assumption.
- C4 — Guarded explanation.** The finite-state workflow and evidence checks keep explanations separate from alert decisions. The campaign did not exercise the LLM path, so it provides no evidence about explanation quality or groundedness in normal operation.
- C5 — Evaluation apparatus.** Replay manifests, isolated subprocesses, repetition plumbing, serialized audit artifacts, the preserved baseline, and the methodology-change log make the results reproducible, including the negative findings.

8.3 Limitations

The paired analysis has only about nine scenario-level observations. The observed no-coordination versus auction effect ($d_z = 0.347$) would need about 65 paired observations for 80% power, or roughly 13 scenarios at the current repetition design. More than twenty implementation parameters influence the results, but the project has no full held-out tuning protocol or seed control.

The metrics also set limits on interpretation. Event-family matching with a ± 5 s tolerance is a system-level measure, not a per-frame detection or tracking measure. The corpus has no per-frame track ground truth, so HOTA, MOTA, and IDF1 cannot be calculated. False alerts per hour are unstable for runs that last only seconds.

This remains a demonstrator-scale evaluation. Only `demo_site_01` is a multi-zone manifest from the campaign; later exploratory multi-zone manifests are outside the 373-run evidence. The CLIP anomaly scorer has an AUC of 0.308 in its calibration note. A post-campaign appearance-descriptor prototype was withheld from the evaluated configuration. It produced no recorded Re-ID matches and did not persist Re-ID features, so the thesis keeps the stated exclusion of person re-identification.

8.4 Future Work

The auction requires field-of-view overlap derived from camera calibration or homography, followed by a comparison with a paced centralized control and stronger allocation baselines such as Hungarian assignment. Verification should examine evidence from the same time as the hypothesis. Fusion should pool dependent observations before combining them.

The scenario corpus should grow to at least thirteen independent scenarios, with controlled seeds, held-out parameter selection, and richer annotations. Per-frame track ground truth would permit HOTA, MOTA, and IDF1. A physical deployment could then measure MQTT/Redis transport cost, edge resources, and operator response instead of relying on in-process replay.

The explanation pipeline needs a campaign with `--llm` enabled, the judge scaffolding executed, and an operator study that separates groundedness from usefulness. A hash chain or signed audit store would make the present append-only log tamper-evident.

8.5 Closing Remarks

The campaign did not confirm the strongest claims from the first demonstration. It did leave a traceable record of why they failed, what was measured, and what should be tested next. That

record is necessary before the prototype can support stronger claims.

Appendix A

Correction Ledger

This appendix preserves a visible record of corrections made during the final evidence audit. It is intentionally included rather than silently replacing earlier claims: the first evaluation was a single-run demonstration, whereas the final campaign consists of repeated replay runs and inferential analysis.

Table A.1: Correction ledger: initial claim, corrected statement, and evidence.

Location	Earlier claim	Corrected statement and evidence
Abstracts; Chapter 7	MAS is approximately 36–38% faster to alert.	The 80.4% reduction occurs only on <code>demo_site_01</code> ($n = 5$, $p = 0.0625$). Across the seven scenarios having all four modes the reduction is 10.7% ($p = 0.59$), not significant. Source: <code>thesis_stats.json</code> .
Abstracts; Chapter 7	Auction coordination halves false alerts and restores precision.	Auction has the lowest marginal mean F1 (0.452). After Holm–Bonferroni correction, none of six pairwise F1 comparisons is significant. Sources: <code>summary_agg.csv</code> ; <code>thesis_stats.json</code> .
Chapter 6, earlier interpretation	“Who verifies” validates the utility-bid design.	Retracted: no campaign manifest populated <code>fov_overlap</code> ; the overlap factor was the 0.5 default. The bid is therefore not a measured view-utility allocation. Sources: <code>camera_agent.py</code> ; coordinator task construction.

Location	Earlier claim	Corrected statement and evidence
Chapter 6, earlier fusion text	Confidence 0.78 demonstrates an audio-video corroboration bonus.	Retracted: the 0.78 alert contains one video event from <code>cam_01</code> . Four of 240 alerts are truly cross-modal, all at 1.0 because noisy-OR saturates. Source: <code>cross_modal_fusion_audit.md</code> .
Chapter 6; Chapter 7	No hallucinated citations were found in replay.	The campaign never invoked <code>--llm</code> ; all 373 explanations used the template. One post-campaign model output was rejected by the guardrail. The claim is a design/property test, not campaign evidence.
Chapter 5	The CLIP anomaly component is presented as working.	Its calibration AUC is 0.308, below chance, consistent with prompt/-domain mismatch. Source: <code>clip_anomaly_calibration_notes.md</code> .
Chapter 6 setup	Variants differ only in architecture.	They also differ in replay pacing and stream sequencing; fusion and cooldown are wall-clock mechanisms. Source: <code>replay.py</code> .
Chapters 1, 5, 7	Approximately 2,500 lines, seven agents, six tests, and a <code>configs/</code> directory.	4,376 tracked Python lines (5,188 including working-tree files); six implemented agents plus an operator console; 26 offline tests; zone data and thresholds are scenario manifests, not a <code>configs/</code> directory.
Chapters 1, 3, 4	No Re-ID.	Retained as the evaluated-system claim. A post-campaign appearance-descriptor prototype was withheld; it did not form part of the reported configuration.
THESIS_REPATCH variance row	Identical auction precision values establish stability.	Rejected: Levene tests give $p = 0.224$ for <code>demo_site_01</code> precision and $p = 0.515$ pooled. Five identical observations do not establish lower variance.

Location	Earlier claim	Corrected statement and evidence
FA/h figures	False alarms per hour is an operational rate.	Over 9–70 second replay runs with 0–3 false positives, FA/h is a scale artefact (maximum 393.9/h), not a deployment-rate estimate.
Chapter 7	The audit stream is immutable.	It is append-only by convention, not cryptographically immutable.

Appendix B

Campaign Scenario Manifests

The following are the nine JSON scenario manifests used by the v2 campaign. They are reproduced directly from the repository so that sensor topology, zone geometry, and ground-truth intervals can be inspected without transcription. The three later exploratory manifests (`loitering_multizone_01`, `wrong_direction_01`, and `zone_occupancy_01`) are not part of the 373-run campaign and are therefore not reported as campaign evidence.

B.1 Abandoned-object scenario

```
{
  "name": "abandoned_object_01",
  "duration_seconds": 78,
  "sensors": [
    {"type": "camera", "id": "cam_01", "source": "data/clips_real/abandoned_object/video1.avi",
     "zones": []}
  ],
  "ground_truth": [
    {"event_type": "abandoned_object", "zone": null, "t_start": 58.0, "t_end": 73.0}
  ],
  "notes": "Source: ABODA video1.avi (see data/clips_real/manifest.json). No shipped frame-level GT. First pass used aura_mas/s"
}
```

B.2 Clock-alarm scenario

```
{
  "name": "audio_alarm_clock_01",
  "duration_seconds": 21,
  "sensors": [
    {"type": "audio", "id": "mic_01", "source": "data/clips_real/audio/clock_alarm_with_baseline.wav"}
  ],
  "ground_truth": [
    {"event_type": "audio_alarm", "zone": null, "t_start": 15.0, "t_end": 20.0}
  ],
  "notes": "Audio-only, added on the YAMNet-integration pass (see results/methodology_changes.md) alongside audio_alarm_siren_0"
}
```

B.3 Siren-alarm scenario

```
{
  "name": "audio_alarm_siren_01",
  "duration_seconds": 21,
  "sensors": [
    {"type": "audio", "id": "mic_01", "source": "data/clips_real/audio/siren_with_baseline.wav"}
  ]
}
```

```

],
"ground_truth": [
  {"event_type": "audio_alarm", "zone": null, "t_start": 15.0, "t_end": 20.0}
],
"notes": "Audio-only, added on the YAMNet-integration pass (see results/methodology_changes.md) to give the DSP-vs-YAMNet abl
}

```

B.4 Glass-break scenario

```

{
  "name": "audio_glass_break_01",
  "duration_seconds": 21,
  "sensors": [
    {"type": "audio", "id": "mic_01", "source": "data/clips_real/audio/glass_breaking_with_baseline.wav"}
  ],
  "ground_truth": [
    {"event_type": "audio_glass_break", "zone": null, "t_start": 15.0, "t_end": 19.5}
  ],
  "notes": "Audio-only, no camera sensor: isolates the audio branch (W1.5) from vision so its recall/false-alert behavior can b
}

```

B.5 Combined audio–video scenario

```

{
  "name": "combined_audio_video_01",
  "duration_seconds": 60,
  "sensors": [
    {"type": "camera", "id": "cam_01", "source": "data/clips/people.mp4",
     "zones": [{"name": "zone_A", "type": "restricted",
                  "polygon": [[0, 60], [768, 60], [768, 432], [0, 432]]}],
    {"type": "audio", "id": "mic_01", "source": "data/clips_real/audio/glass_breaking_with_baseline.wav", "zone": "zone_A"}
  ],
  "ground_truth": [
    {"event_type": "intrusion", "zone": "zone_A", "t_start": 15.0, "t_end": 34.0},
    {"event_type": "audio_glass_break", "zone": "zone_A", "t_start": 15.0, "t_end": 19.5}
  ],
  "notes": "cam_01 originally pointed at data/clips/intrusion.mp4, a synthetic vector-graphic placeholder (not real footage) th
}

```

B.6 Demonstrator site scenario

```

{
  "name": "demo_site_01",
  "duration_seconds": 55,
  "sensors": [
    {"type": "camera", "id": "cam_01", "source": "data/clips/people.mp4",
     "zones": [
       {"name": "zone_A", "type": "restricted",
        "polygon": [[380, 220], [768, 220], [768, 432], [380, 432]]},
       {"name": "entry", "type": "entry",
        "polygon": [[0, 300], [380, 300], [380, 432], [0, 432]]}
     ]},
    {"type": "camera", "id": "cam_02", "source": "data/clips/street.mp4",
     "zones": [
       {"name": "zone_B", "type": "restricted",
        "polygon": [[0, 400], [960, 400], [960, 540], [0, 540]]}
     ]},
    {"type": "audio", "id": "mic_01", "source": "data/clips/glass_break.wav"}
  ],
  "ground_truth": [
    {"event_type": "intrusion", "zone": "zone_A", "t_start": 3.0, "t_end": 35.0},
    {"event_type": "loitering", "zone": "entry", "t_start": 16.0, "t_end": 46.0},
    {"event_type": "audio_glass_break", "zone": null, "t_start": 14.0, "t_end": 16.0}
  ]
}

```

```
]
}
```

B.7 Fight scenario

```
{
  "name": "fight_01",
  "duration_seconds": 8,
  "sensors": [
    {"type": "camera", "id": "cam_01", "source": "data/clips_real/violence/violent_1.mp4",
     "zones": [], "enable_clip": true}
  ],
  "ground_truth": [
    {"event_type": "anomaly", "zone": null, "t_start": 0.5, "t_end": 4.5}
  ],
  "notes": "Zone rules (intrusion/loitering/abandoned_object) cannot detect this - 'people fighting violently' is only reachable"
}
```

B.8 Intrusion scenario

```
{
  "name": "intrusion_01",
  "duration_seconds": 55,
  "sensors": [
    {"type": "camera", "id": "cam_01", "source": "data/clips/people.mp4",
     "zones": [{"name": "zone_A", "type": "restricted",
                  "polygon": [[0, 60], [768, 60], [768, 432], [0, 432]]}],
     {"type": "camera", "id": "cam_02", "source": "data/clips/street.mp4",
     "zones": []},
     {"type": "audio", "id": "mic_01", "source": "data/clips/glass_break.wav"}
  ],
  "ground_truth": [
    {"event_type": "intrusion", "zone": "zone_A", "t_start": 15.0, "t_end": 34.0},
    {"event_type": "audio_glass_break", "zone": null, "t_start": 14.0, "t_end": 16.0}
  ],
  "notes": "cam_01/cam_02 originally pointed at data/clips/intrusion.mp4 and overview.mp4 -- both turned out to be synthetic videos"
}
```

B.9 Loitering true-negative probe

```
{
  "name": "loitering_01",
  "duration_seconds": 55,
  "sensors": [
    {"type": "camera", "id": "cam_01", "source": "data/clips/people.mp4",
     "zones": [
       {"name": "entry", "type": "entry",
        "polygon": [[0, 300], [380, 300], [380, 432], [0, 432]]}
     ]
  ],
  "ground_truth": [],
  "notes": "Deliberately EMPTY ground truth. The original demo_site_01.json (pre-existing in this repo) claimed a loitering GT"
}
```

Appendix C

Data Provenance and Licence Notes

Table C.1 transcribes the provenance information in `data/clips_real/manifest.json`. These clips extend the original demonstrator assets; they are used only for non-commercial academic evaluation.

Table C.1: Provenance and licence notes for real evaluation assets.

Asset	Source	Licence / use note	Role in corpus
violence/violent_brawl.mp4	Blanco et al. [85]	Free for re-search and educational purposes	Simulated brawl, about 4.8 s, 1920×1080 at 30 FPS.
violence/violent_shooting.mp4	Same dataset	Free for re-search and educational purposes	Available corpus asset; not a separately reported manifest.
violence/nonviolent_shooting.mp4	Same dataset	Free for re-search and educational purposes	Deliberately ambiguous non-violent hard negative.
abandoned_object_ABODA.avi	ABODA [87]	Public research dataset; no explicit per-file licence in its repository	73-second indoor hallway sequence. Its onset is an approximate, frame-checked annotation because ABODA supplies no frame-level ground truth.
audio/glass_breaking_esc50.wav	ESC-50 [86]	CC BY-NC 3.0	Real five-second glass-breaking recording.
audio/clock_alarm_esc50.wav	ESC-50 [86]	CC BY-NC 3.0	Alarm proxy event.
audio/siren_esc50.wav	ESC-50 [86]	CC BY-NC 3.0	Five-second siren hazard event.

Appendix D

Configuration and Measurement Parameters

The reported scores are outputs of a system with numerous free parameters. Table D.1 makes the principal defaults explicit, including their implementation provenance. It is a disclosure table, not a tuning study: apart from the documented CLIP threshold sweep, these values were not selected on a held-out validation split.

Table D.1: Principal free parameters and source locations.

Parameter		Default / campaign value	Implementation provenance
Fusion window		6 s	<code>fusion_agent.py</code> , <code>FusionAgent.window_seconds</code>
Video	reliability	0.9	<code>fusion_agent.py</code> , noisy-OR weighting
Audio	reliability	0.7	<code>fusion_agent.py</code> , noisy-OR weighting
Cross-modality bonus		0.05	<code>fusion_agent.py</code> , confidence ad- justment
Cross-sensor bonus		0.05	<code>fusion_agent.py</code> , confidence ad- justment
Verification gray zone		[0.35, 0.75)	<code>coordinator_agent.py</code> , <code>needs_verification()</code>
Bid window		1.0 s	<code>coordinator_agent.py</code> , construc- tor default
Verification timeout		3.0 s	<code>coordinator_agent.py</code> , construc- tor default
Origin-sensor bid fac- tor		0.3	<code>camera_agent.py</code> , <code>_view_score()</code>
Busy-camera capacity factor		0.2	<code>camera_agent.py</code> , <code>_view_score()</code>
FoV overlap fallback		0.5	<code>camera_agent.py</code> : task default; in- active as an information-bearing term in campaign measurement
Critical / warning / info thresholds		0.45 / 0.55 / 0.70	<code>policy_agent.py</code> , <code>ALERT_THRESHOLDS</code>

Parameter		Default / campaign value	Implementation provenance
Verification adjust- ment		+0.15 / - 0.20	policy_agent.py, verification out- come handling
Alert cooldown		20 s	policy_agent.py, constructor de- fault
Inference cadence		5 FPS	camera_agent.py, infer_fps de- fault
Camera confidence floor		0.35	camera_agent.py, detector config- uration
Verification confi- dence / size		0.25 / 960 px	camera_agent.py, _verify()
CLIP anomaly thresh- old		0.55	camera_agent.py, constructor de- fault
Loitering threshold		8 s	camera_agent.py, ZoneRuleEngine default
Abandoned-object threshold		10 s	camera_agent.py, ZoneRuleEngine default
Static-object IoU		0.6	camera_agent.py, ZoneRuleEngine default
Metric time tolerance		± 5 s	metrics.py, event matching logic

Appendix E

Methodology Changes from v1 to v2

This table is reproduced from `results/methodology_changes.md`. It records changes that could move a result relative to the preserved pre-YAMNet baseline, avoiding the appearance that v1 and v2 scores measure an unchanged pipeline.

Table E.1: Documented changes between the v1 and v2 evaluation campaigns.

#	Change	Why	Expected direction
1	Local YAMNet integration instead of DSP fallback only	The TensorFlow Hub URL was dead; this blocked typed audio events.	Audio-scenario recall and precision up.
2	Lookback-extended window and max pooling instead of bare one-second mean pooling	Mean pooling diluted brief transients (0.069 versus 0.75 on the same clip).	Audio recall up.
3	One event per type per audio chunk	Multiple matched classes otherwise entered noisy-OR as independent evidence.	Inflated confidence and false positives down.
4	Unified event-family mapping	Metric and fusion family dictionaries had drifted.	Correctly mapped audio events can score as matches.
5	Audio events carry the scenario zone	Without the zone, audio and video could not join the same fusion hypothesis.	Confidence can rise only for time- and zone-aligned evidence.
6	Scenario clock starts after model warm-up	Cold-start loading had been counted in time-to-alert.	Wall seconds and apparent alert delay down.
7	Repetition flags and rep-suffixed outputs	Reruns previously overwrote results.	Enables mean and standard-deviation reporting.

#	Change	Why	Expected direction
8	Aggregation to summary_agg.csv	Repeated rows needed a separate aggregation artifact.	No direct scoring effect.
9	Two additional ESC-50 alarm scenarios	Earlier audio sce- narios exercised only glass-break ground truth.	Expands class- discrimination cov- erage.
10	Full nine-scenario, four-mode, five- repetition cam- paign	Single-run demon- stration could not support uncertainty estimates.	Enables paired inferen- tial analysis.

Appendix F

Message Schemas

The implementation uses typed data classes in `aura_mas/core/bus.py`. The abbreviated field inventory below distinguishes transient detections from durable semantic events and operator-facing alerts.

Table F.1: Core message-schema fields.

Schema	Key fields
Detection	<code>sensor_id</code> , <code>frame_id</code> , <code>timestamp</code> , object class, confidence, bounding box, and track identifier.
Event	<code>event_id</code> , <code>sensor_id</code> , <code>timestamp</code> , <code>event_type</code> , confidence, modality, zone, track identifier, evidence path, and extensible metadata.
Alert	<code>alert_id</code> , <code>timestamp</code> , severity, event type, confidence, zone, sensor list, evidence list, fused-event list, explanation, and status.

The topic constants in the same module route detections, events, coordination tasks, bids, awards, verifications, alerts, and audit records. This explicit separation makes it possible to exchange structured facts rather than raw footage across the system boundary.

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